



AWS CHEAT SHEETS

The following is a compilation of the most relevant AWS services cheat sheets, which are among the core topics in the Solutions Architect Professional Exam. Head over to the [Tutorials Dojo website](https://portal.tutorialsdojo.com/) to view our complete library of AWS cheat sheets.

Amazon VPC

- Create a virtual network in the cloud dedicated to your AWS account where you can launch AWS resources
- Amazon VPC is the networking layer of Amazon EC2
- A VPC spans all the Availability Zones in the region. After creating a VPC, you can add one or more subnets in each Availability Zone.

Key Concepts

- A **virtual private cloud** (VPC) allows you to specify an IP address range for the VPC, add subnets, associate security groups, and configure route tables.
- A **subnet** is a range of IP addresses in your VPC. You can launch AWS resources into a specified subnet. Use a **public subnet** for resources that must be connected to the internet, and a **private subnet** for resources that won't be connected to the internet.
- To protect the AWS resources in each subnet, use **security groups** and **network access control lists (ACL)**.
- Expand your VPC by adding secondary IP ranges.



Default vs Non-Default VPC

Default VPC	Non-Default VPC
New AWS accounts comes with a default VPC that has a default subnet in each Availability Zone.	You can create your own non-default VPC, and configure it as you need. Subnets that you create in your non-default VPC and additional subnets that you create in your default VPC are called non-default subnets.
Pre-configured with a default set of subnets, security groups, and route tables.	Must be configured by the user, including subnets, security groups, and route tables.
Your default VPC includes an internet gateway, which allows your instances to communicate with the internet, and each default subnet is a public subnet.	Instances can communicate with each other, but can't access the internet. You can enable internet access for an instance launched into a non-default subnet by attaching an Internet Gateway and associating an Elastic IP address with the instance.
AWS automatically assigned a public IP address to instances launched in the default subnet, unless specified otherwise.	By default, each instance that you launch into a non-default subnet has a private IPv4 address, but no public IPv4 address, unless you specifically assign one at launch, or you modify the subnet's public IP address attribute.
To allow an instance in your VPC to initiate outbound connections to the internet but prevent unsolicited inbound connections from the internet, you can use a network address translation (NAT) device or NAT Gateway for IPv4 traffic.	
You can optionally associate an Amazon-provided IPv6 CIDR block with your VPC and assign IPv6 addresses to your instances. IPv6 traffic is separate from IPv4 traffic; your route tables must include separate routes for IPv6 traffic.	

Accessing a Corporate or Home Network

- You can optionally connect your VPC to your own corporate data center using an **IPsec AWS managed VPN connection**, making the AWS Cloud an extension of your data center.
- A **VPN connection** consists of:
 - a **virtual private gateway** (which is the VPN concentrator on the Amazon side of the VPN connection) attached to your VPC.



- a **customer gateway** (which is a physical device or software appliance on your side of the VPN connection) located in your data center.
- **AWS Site-to-Site Virtual Private Network (VPN)** connections can be moved from a virtual private gateway to an **AWS Transit Gateway** without having to make any changes on your customer gateway. Transit Gateways enable you to easily scale connectivity across thousands of Amazon VPCs, AWS accounts, and on-premises networks.
- **AWS PrivateLink** enables you to privately connect your VPC to supported AWS services, services hosted by other AWS accounts (VPC endpoint services), and supported AWS Marketplace partner services. You do not require an internet gateway, NAT device, public IP address, AWS Direct Connect connection, or VPN connection to communicate with the service. Traffic between your VPC and the service does not leave the Amazon network.
- You can create a **VPC peering connection** between your VPCs, or with a VPC in another AWS account, and enable routing of traffic between the VPCs using private IP addresses. You cannot create a VPC peering connection between VPCs that have overlapping CIDR blocks.
- Applications in an Amazon VPC can securely access AWS PrivateLink endpoints across VPC peering connections. The support of VPC peering by AWS PrivateLink makes it possible for customers to privately connect to a service even if that service's endpoint resides in a different Amazon VPC that is connected using VPC peering.
- AWS PrivateLink endpoints can now be accessed across both intra- and inter-region VPC peering connections.

VPC Use Case Scenarios

- VPC with a Single Public Subnet
- VPC with Public and Private Subnets (NAT)
- VPC with Public and Private Subnets and AWS Managed VPN Access
- VPC with a Private Subnet Only and AWS Managed VPN Access

Subnets

- When you create a VPC, you must specify a range of IPv4 addresses for the VPC in the form of a Classless Inter-Domain Routing (CIDR) block (example: 10.0.0.0/16). This is the **primary CIDR block** for your VPC.
- You can add one or more subnets in each Availability Zone of your VPC's region.
- You specify the CIDR block for a subnet, which is a subset of the VPC CIDR block.
- A CIDR block must not overlap with any existing CIDR block that's associated with the VPC.
- Types of Subnets
 - Public Subnet - has an internet gateway
 - Private Subnet - doesn't have an internet gateway



- VPN-only Subnet - has a virtual private gateway instead
- IPv4 CIDR block size should be between a /16 netmask (65,536 IP addresses) and /28 netmask (16 IP addresses).
- The **first four IP addresses and the last IP address in each subnet CIDR block** are **NOT available** for you to use, and cannot be assigned to an instance.
- You cannot increase or decrease the size of an existing CIDR block.
- When you associate a CIDR block with your VPC, a route is automatically added to your VPC route tables to enable routing within the VPC (the destination is the CIDR block and the target is *local*).
- You have a limit on the number of CIDR blocks you can associate with a VPC and the number of routes you can add to a route table.
- The following rules apply when you add IPv4 CIDR blocks to a VPC that's part of a **VPC peering connection**:
 - If the VPC peering connection is active, you can add CIDR blocks to a VPC provided they do not overlap with a CIDR block of the peer VPC.
 - If the VPC peering connection is pending-acceptance, the owner of the requester VPC cannot add any CIDR block to the VPC. Either the owner of the acceptor VPC must accept the peering connection, or the owner of the requester VPC must delete the VPC peering connection request, add the CIDR block, and then request a new VPC peering connection.
 - If the VPC peering connection is pending-acceptance, the owner of the acceptor VPC can add CIDR blocks to the VPC. If a secondary CIDR block overlaps with a CIDR block of the requester VPC, the VPC peering connection request fails and cannot be accepted.
- If you're using AWS Direct Connect to connect to multiple VPCs through a direct connect gateway, the VPCs that are associated with the direct connect gateway must not have overlapping CIDR blocks.
- The CIDR block is ready for you to use when it's in the *associated* state.
- You can disassociate a CIDR block that you've associated with your VPC; however, you cannot disassociate the primary CIDR block.

Subnet Routing

- Each subnet must be associated with a **route table**, which specifies the allowed routes for **outbound traffic** leaving the subnet.
- Every subnet that you create is automatically associated with the main route table for the VPC.
- You can change the association, and you can change the contents of the main route table.
- You can allow an instance in your VPC to initiate outbound connections to the internet over IPv4 but prevent unsolicited inbound connections from the internet using a **NAT gateway or NAT instance**.
- To initiate outbound-only communication to the internet over IPv6, you can use an egress-only internet gateway.

Subnet Security

- Security Groups — control inbound and outbound traffic for your instances
 - You can associate one or more (up to five) security groups to an instance in your VPC.



- If you don't specify a security group, the instance automatically belongs to the default security group.
- When you create a security group, it has no inbound rules. By default, it includes an outbound rule that allows all outbound traffic.
- Security groups are associated with network interfaces.
- Network Access Control Lists – control inbound and outbound traffic for your subnets
 - Each subnet in your VPC must be associated with a network ACL. If none is associated, automatically associated with the default network ACL.
 - You can associate a network ACL with multiple subnets; however, a subnet can be associated with only one network ACL at a time.
 - A network ACL contains a numbered list of rules that is evaluated in order, starting with the lowest numbered rule, to determine whether traffic is allowed in or out of any subnet associated with the network ACL.
 - The default network ACL is configured to **allow all traffic to flow in and out** of the subnets to which it is associated.
 - For custom ACLs, you need to add a rule for ephemeral ports, usually with the range of 32768-65535. If you have a NAT Gateway, ELB or a Lambda function in a VPC, you need to enable 1024-65535 port range.
- Flow logs – capture information about the IP traffic going to and from network interfaces in your VPC that is published to CloudWatch Logs.
- Flow logs can help you with a number of tasks, such as:
 - Diagnosing overly restrictive security group rules
 - Monitoring the traffic that is reaching your instance
 - Determining the direction of the traffic to and from the network interfaces
- Flow log data is collected outside of the path of your network traffic, and therefore does not affect network throughput or latency. You can create or delete flow logs without any risk of impact to network performance.
- After you've created a flow log, it can take several minutes to begin collecting and publishing data to the chosen destinations. Flow logs do not capture real-time log streams for your network interfaces.
- VPC Flow Logs can be sent directly to an Amazon S3 bucket which allows you to retrieve and analyze these logs yourself.
- Amazon security groups and network ACLs don't filter traffic to or from link-local addresses or AWS-reserved IPv4 addresses. Flow logs do not capture IP traffic to or from these addresses.



Security Group vs NACL

Security Group	Network Access Control List
Acts as a firewall for associated Amazon EC2 instances.	Acts as a firewall for associated subnets.
Controls both inbound and outbound traffic at the instance level.	Controls both inbound and outbound traffic at the subnet level.
You can secure your VPC instances using only security groups.	Network ACLs are an additional layer of defense.
Supports allow rules only.	Supports allow rules and deny rules.
Stateful (Return traffic is automatically allowed, regardless of any rules).	Stateless (Return traffic must be explicitly allowed by rules).
Evaluates all rules before deciding whether to allow traffic.	Evaluates rules in number order when deciding whether to allow traffic, starting with the lowest-numbered rule.
Applies only to the instance that is associated with it.	Applies to all instances in the subnet it is associated with.
Has separate rules for inbound and outbound traffic.	Has separate rules for inbound and outbound traffic.
A newly created security group denies all inbound traffic by default.	A newly created nACL denies all inbound traffic by default.
A newly created security group has an outbound rule that allows all outbound traffic by default.	A newly created nACL denies all outbound traffic by default.
Instances associated with a security group can't talk to each other unless you add rules allowing it.	Each subnet in your VPC must be associated with a network ACL. If none is associated, the default nACL is selected.
Security groups are associated with network interfaces.	You can associate a network ACL with multiple subnets; however, a subnet can be associated with only one network ACL at a time.



VPC Networking Components

- Network Interfaces
 - a virtual network interface that can include:
 - a primary private IPv4 address
 - one or more secondary private IPv4 addresses
 - one Elastic IP address per private IPv4 address
 - one public IPv4 address, which can be auto-assigned to the network interface for eth0 when you launch an instance
 - one or more IPv6 addresses
 - one or more security groups
 - a MAC address
 - a source/destination check flag
 - a description
 - Network interfaces can be attached and detached from instances, however, you cannot detach a primary network interface.
- Route Tables
 - contains a set of rules, called *routes*, that are used to determine where network traffic is directed.
 - A subnet can only be associated with one route table at a time, but you can associate multiple subnets with the same route table.
 - You cannot delete the main route table, but you can replace the main route table with a custom table that you've created.
 - You must update the route table for any subnet that uses gateways or connections.
 - Uses the most specific route in your route table that matches the traffic to determine how to route the traffic (longest prefix match).
- Internet Gateways
 - Allows communication between instances in your VPC and the internet.
 - Imposes no availability risks or bandwidth constraints on your network traffic.
 - Provides a target in your VPC route tables for internet-routable traffic, and performs network address translation for instances that have been assigned public IPv4 addresses.
 - The following table provides an overview of whether your VPC automatically comes with the components required for internet access over IPv4 or IPv6.
 - To enable access to or from the Internet for instances in a VPC subnet, you must do the following:
 - Attach an Internet Gateway to your VPC
 - Ensure that your subnet's route table points to the Internet Gateway.
 - Ensure that instances in your subnet have a globally unique IP address (public IPv4 address, Elastic IP address, or IPv6 address).

- Ensure that your network access control and security group rules allow the relevant traffic to flow to and from your instance

	Default VPC	Non-default VPC
Internet gateway	Yes	Yes, if you created the VPC using the first or second option in the VPC wizard. Otherwise, you must manually create and attach the internet gateway.
Route table with route to internet gateway for IPv4 traffic (0.0.0.0/0)	Yes	Yes, if you created the VPC using the first or second option in the VPC wizard. Otherwise, you must manually create the route table and add the route.
Route table with route to internet gateway for IPv6 traffic (::/0)	No	Yes, if you created the VPC using the first or second option in the VPC wizard, and if you specified the option to associate an IPv6 CIDR block with the VPC. Otherwise, you must manually create the route table and add the route.
Public IPv4 address automatically assigned to instance launched into subnet	Yes (default subnet)	No (non-default subnet)
IPv6 address automatically assigned to instance launched into subnet	No (default subnet)	No (non-default subnet)

- Egress-Only Internet Gateways
 - VPC component that allows outbound communication over IPv6 from instances in your VPC to the Internet, and prevents the Internet from initiating an IPv6 connection with your instances.
 - An egress-only Internet gateway is stateful.
 - You cannot associate a security group with an egress-only Internet gateway.



- You can use a network ACL to control the traffic to and from the subnet for which the egress-only Internet gateway routes traffic.
- NAT
 - Enable instances in a private subnet to connect to the internet or other AWS services, but prevent the internet from initiating connections with the instances.
 - NAT Gateways
 - You must specify the **public subnet** in which the NAT gateway should reside.
 - You must specify an **Elastic IP address** to associate with the NAT gateway when you create it.
 - Each NAT gateway is created in a specific Availability Zone and implemented with redundancy in that zone.
 - Deleting a NAT gateway disassociates its Elastic IP address, but does not release the address from your account.
 - A NAT gateway supports the following protocols: TCP, UDP, and ICMP.
 - You cannot associate a security group with a NAT gateway.
 - A NAT gateway can support up to 55,000 simultaneous connections to each unique destination.
 - A NAT gateway cannot send traffic over VPC endpoints, VPN connections, AWS Direct Connect, or VPC peering connections.
 - A NAT gateway uses ports 1024-65535. Make sure to enable these in the inbound rules of your network ACL.
 - NAT gateways do not support IPv6 traffic—use an outbound-only (egress-only) internet gateway instead.
 - NAT Instance vs NAT Gateways

Attribute	NAT gateway	NAT instance
Availability	Highly available. NAT gateways in each Availability Zone are implemented with redundancy. Create a NAT gateway in each Availability Zone to ensure zone-independent architecture.	Use a script to manage failover between instances.
Bandwidth	Can scale up to 45 Gbps.	Depends on the bandwidth of the instance type.
Maintenance	Managed by AWS.	Managed by you.
Performance	Software is optimized for handling NAT traffic.	A generic Amazon Linux AMI that's configured to perform NAT.
Cost	Charged depending on the number of NAT gateways you use, duration of usage, and amount of data that you send through the NAT gateways.	Charged depending on the number of NAT instances that you use, duration of usage, and instance type and size.



Type and Size	Uniform offering: you don't need to decide on the type or size.	Choose a suitable instance type and size, according to your predicted workload.
Public IP Addresses	Choose the Elastic IP address to associate with a NAT gateway at creation.	Use an elastic IP address or a public IP address with a NAT instance. You can change the public IP address at any time by associating a new elastic IP address with the instance.
Private IP Addresses	Automatically selected from the subnet's IP address range when you create the gateway.	Assign a specific private IP address from the subnet's IP address range when you launch the instance.
Security Groups	Cannot be associated with a NAT gateway.	Associate with your NAT instance and the resources behind your NAT instance to control inbound and outbound traffic.
Network ACLs	Use a network ACL to control the traffic to and from the subnet in which your NAT gateway resides.	Use a network ACL to control the traffic to and from the subnet in which your NAT instance resides.
Flow Logs	Use flow logs to capture the traffic.	Use flow logs to capture the traffic.
Port Forwarding	Not supported.	Manually customize the configuration to support port forwarding.
Bastion Servers	Not supported.	Use as a bastion server.
Traffic Metrics	Monitor your NAT gateway using CloudWatch.	View CloudWatch metrics for the instance.
Timeout Behavior	When a connection times out, a NAT gateway returns an RST packet to any resources behind the NAT gateway that attempt to continue the connection (it does not send a FIN packet).	When a connection times out, a NAT instance sends a FIN packet to the resources behind the NAT instance to close the connection.
IP Fragmentation	Supports forwarding of IP fragmented packets for the UDP protocol. Does not support fragmentation for the TCP and ICMP protocols. Fragmented packets for these protocols will get dropped.	Supports reassembly of IP fragmented packets for the UDP, TCP, and ICMP protocols.

- DHCP Options Sets



- **Dynamic Host Configuration Protocol (DHCP)** provides a standard for passing configuration information to hosts on a TCP/IP network.
- You can assign your own domain name to your instances, and use up to four of your own DNS servers by specifying a special set of DHCP options to use with the VPC.
- Creating a VPC automatically creates a set of DHCP options, which are domain-name-servers=AmazonProvidedDNS, and domain-name=domain-name-for-your-region, and associates them with the VPC.
- After you create a set of DHCP options, you can't modify them. Create a new set and associate a different set of DHCP options with your VPC, or use no DHCP options at all.
- **DNS**
 - AWS provides instances launched in a default VPC with public and private DNS hostnames that correspond to the public IPv4 and private IPv4 addresses for the instance.
 - AWS provides instances launched in a non-default VPC with private DNS hostname and possibly a public DNS hostname, depending on the DNS attributes you specify for the VPC and if your instance has a public IPv4 address.
 - Set VPC attributes *enableDnsHostnames* and *enableDnsSupport* to true so that your instances receive a public DNS hostname and Amazon-provided DNS server can resolve Amazon-provided private DNS hostnames.
 - If you use custom DNS domain names defined in a private hosted zone in Route 53, the *enableDnsHostnames* and *enableDnsSupport* attributes must be set to true.
- **VPC Peering**
 - A networking connection between two VPCs that enables you to route traffic between them privately. Instances in either VPC can communicate with each other as if they are within the same network.
- **Elastic IP Addresses**
 - **A static, public IPv4 address.**
 - You can associate an Elastic IP address with any instance or network interface for any VPC in your account.
 - You can mask the failure of an instance by rapidly remapping the address to another instance in your VPC.
 - Your Elastic IP addresses remain associated with your AWS account until you explicitly release them.
 - AWS imposes a small hourly charge when EIPs aren't associated with a running instance, or when they are associated with a stopped instance or an unattached network interface.
 - You're limited to five Elastic IP addresses.
- **VPC Endpoints**
 - Privately connect your VPC to supported AWS services and VPC endpoint services powered by PrivateLink without requiring an internet gateway, NAT device, VPN connection, or AWS Direct Connect connection.
 - Endpoints are virtual devices.
 - Two Types



■ Interface Endpoints

- An elastic network interface with a private IP address that serves as an entry point for traffic destined to a supported service.
- Can be accessed through AWS VPN connections or AWS Direct Connect connections, through intra-region VPC peering connections from Nitro instances, and through inter-region VPC peering connections from any type of instance.
- For each interface endpoint, you can choose only one subnet per Availability Zone. Endpoints are supported within the same region only.
- Interface endpoints do not support the use of endpoint policies.
- An interface endpoint supports IPv4 TCP traffic only.

Gateway Endpoints

- A gateway that is a target for a specified route in your route table, used for traffic destined to a supported AWS service.
- You can create multiple endpoints in a single VPC, for example, to multiple services. You can also create multiple endpoints for a single service, and use different route tables to enforce different access policies from different subnets to the same service.
- You can modify the endpoint policy that's attached to your endpoint, and add or remove the route tables that are used by the endpoint.
- Endpoints are supported within the same region only. You cannot create an endpoint between a VPC and a service in a different region.
- Endpoints support IPv4 traffic only.
- You must enable DNS resolution in your VPC, or if you're using your own DNS server, ensure that DNS requests to the required service (such as S3) are resolved correctly to the IP addresses maintained by AWS.

You can create your own application in your VPC and configure it as an AWS PrivateLink-powered service (referred to as an *endpoint service*). You are the *service provider*, and the AWS principals that create connections to your service are *service consumers*.

VPN Connections

VPN connectivity option	Description
AWS managed VPN	You can create an IPsec VPN connection between your VPC and your remote network. On the AWS side of the VPN connection, a <i>virtual private gateway</i> provides two VPN endpoints (tunnels) for automatic failover. You



	configure your <i>customer gateway</i> on the remote side of the VPN connection.
AWS VPN CloudHub	If you have more than one remote network, you can create multiple AWS managed VPN connections via your virtual private gateway to enable communication between these networks.
Third party software VPN appliance	You can create a VPN connection to your remote network by using an Amazon EC2 instance in your VPC that's running a third party software VPN appliance. AWS does not provide or maintain third party software VPN appliances; however, you can choose from a range of products provided by partners and open source communities.
AWS Direct Connect	You can also use AWS Direct Connect to create a dedicated private connection from a remote network to your VPC. You can combine this connection with an AWS managed VPN connection to create an IPsec-encrypted connection.

- Specify a private Autonomous System Number (ASN) for the virtual private gateway. If you don't specify an ASN, the virtual private gateway is created with the default ASN (64512). You cannot change the ASN after you've created the virtual private gateway.
- When you create a VPN connection, you must:
 - Specify the type of routing that you plan to use (static or dynamic)
 - Update the route table for your subnet
- If your VPN device supports Border Gateway Protocol (BGP), specify **dynamic routing** when you configure your VPN connection. If your device does not support BGP, specify **static routing**.
- VPG uses path selection to determine how to route traffic to your remote network. Longest prefix match applies.
- Each VPN connection has two tunnels, with each tunnel using a unique virtual private gateway public IP address. It is important to configure both tunnels for redundancy.

Pricing



- Charged for VPN Connection-hour
- Charged for each "NAT Gateway-hour" that your NAT gateway is provisioned and available.
- Data processing charges apply for each Gigabyte processed through the NAT gateway regardless of the traffic's source or destination.
- You also incur standard AWS data transfer charges for all data transferred via the NAT gateway.
- Charges for unused or inactive Elastic IPs.

References:

<https://docs.aws.amazon.com/vpc/latest/userguide/what-is-amazon-vpc.html>

<https://aws.amazon.com/vpc/details/>

<https://aws.amazon.com/vpc/pricing/>

<https://aws.amazon.com/vpc/faqs/>

Amazon CloudFront

- A web service that speeds up distribution of your static and dynamic web content to your users. A Content Delivery Network (CDN) service.
- It delivers your content through a worldwide network of data centers called **edge locations**. When a user requests content that you're serving with CloudFront, the user is routed to the edge location that provides the lowest latency, so that content is delivered with the best possible performance.
 - If the content is already in the edge location with the lowest latency, CloudFront delivers it immediately.
 - If the content is not in that edge location, CloudFront retrieves it from an origin that you've defined
- **How CloudFront Delivers Content**
 - You specify **origin servers**, like an S3 bucket or your own HTTP server, from which CloudFront gets your files which will then be distributed from CloudFront edge locations all over the world.
 - Upload your files to your origin servers. Your files, also known as **objects**.
 - Create a **CloudFront distribution**, which tells CloudFront which origin servers to get your files from when users request the files through your web site or application. At the same time, you specify details such as whether you want CloudFront to log all requests and whether you want the distribution to be enabled as soon as it's created.
 - CloudFront assigns a domain name to your new distribution that you can see in the CloudFront console.
 - CloudFront sends your distribution's configuration (but not your content) to all of its **edge locations**—collections of servers in geographically dispersed data centers where CloudFront caches copies of your objects.
- CloudFront supports the **WebSocket protocol** as well as the **HTTP protocol** with the following HTTP methods:
 - GET



- HEAD
- POST
- PUT
- DELETE
- OPTIONS
- PATCH.
- Using **Lambda@Edge** with CloudFront enables a variety of ways to customize the content that CloudFront delivers. It can help you configure your CloudFront distribution to serve private content from your own custom origin, as an option to using signed URLs or signed cookies.(See AWS Compute Services Lambda Lambda@Edge)
- CloudFront also has **regional edge caches** that bring more of your content closer to your viewers, even when the content is not popular enough to stay at a CloudFront edge location, to help improve performance for that content.
- You can use a zone apex name on CloudFront
- CloudFront supports wildcard CNAME
- Different CloudFront Origins
 - **Using S3 buckets for your origin** - you place any objects that you want CloudFront to deliver in an S3 bucket.
 - **Using S3 buckets configured as website endpoints for your origin**
 - **Using a mediastore container or a media package channel for your origin** - you can set up an S3 bucket that is configured as a MediaStore container, or create a channel and endpoints with MediaPackage. Then you create and configure a distribution in CloudFront to stream the video.
 - **Using EC2 or other custom origins** - A custom origin is an HTTP server, for example, a web server.
 - **Using CloudFront Origin Groups for origin failover** - use origin failover to designate a primary origin for CloudFront plus a second origin that CloudFront automatically switches to when the primary origin returns specific HTTP status code failure responses.
- Objects are cached for 24 hours by default. You can invalidate files in CloudFront edge caches even before they expire.
- You can configure CloudFront to automatically compress files of certain types and serve the compressed files when viewer requests include *Accept-Encoding: gzip* in the request header.
- CloudFront can cache different versions of your content based on the values of query string parameters.
- CloudFront Distributions
 - You create a **CloudFront distribution** to tell CloudFront where you want content to be delivered from, and the details about how to track and manage content delivery.
 - You create a distribution and choose the configuration settings you want:
 - Your content origin—that is, the Amazon S3 bucket, MediaPackage channel, or HTTP server from which CloudFront gets the files to distribute. You can specify any combination of up to 25 S3 buckets, channels, and/or HTTP servers as your origins.



- Access—whether you want the files to be available to everyone or restrict access to some users.
- Security—whether you want CloudFront to require users to use HTTPS to access your content.
- Cookie or query-string forwarding—whether you want CloudFront to forward cookies or query strings to your origin.
- Geo-restrictions—whether you want CloudFront to prevent users in selected countries from accessing your content.
- Access logs—whether you want CloudFront to create access logs that show viewer activity.
- You can use distributions to serve the following content over HTTP or HTTPS:
 - Static and dynamic download content.
 - Video on demand in different formats, such as Apple HTTP Live Streaming (HLS) and Microsoft Smooth Streaming.
 - A live event, such as a meeting, conference, or concert, in real time.
- Values that you specify when you create or update a distribution
 - Delivery Method - Web or RTMP.
 - Origin Settings - information about one or more locations where you store the original versions of your web content.
 - Cache Behavior Settings - lets you configure a variety of CloudFront functionality for a given URL path pattern for files on your website.
 - Custom Error Pages and Error Caching
 - Restrictions - if you need to prevent users in selected countries from accessing your content, you can configure your CloudFront distribution either to allow users in a **whitelist** of specified countries to access your content or to not allow users in a **blacklist** of specified countries to access your content.
- **Cache Behavior Settings**
 - The functionality that you can configure for each cache behavior includes:
 - The path pattern.
 - If you have configured multiple origins for your CloudFront distribution, which origin you want CloudFront to forward your requests to.
 - Whether to forward query strings to your origin.
 - Whether accessing the specified files requires signed URLs.
 - Whether to require users to use HTTPS to access those files.
 - The minimum amount of time that those files stay in the CloudFront cache regardless of the value of any Cache-Control headers that your origin adds to the files.
 - After creating your CloudFront distribution, you can invalidate its cached items by creating an invalidation request.
- **Price Class**



- Choose the price class that corresponds with the maximum price that you want to pay for CloudFront service. By default, CloudFront serves your objects from edge locations in all CloudFront regions.
- **Performance and Availability**
 - CloudFront also allows you to set up multiple origins to enable redundancy with **Origin Failover**. To set up origin failover, you must have a distribution with at least two origins. Next, you create an origin group for your distribution that includes the two origins, setting one as the primary. Finally, you define a cache behavior in which you specify the origin group as your origin.
 - The two origins in the origin group can be any combination of the following: AWS origins, like Amazon S3 buckets or Amazon EC2 instances, or custom origins, like your own HTTP web server.
 - When you create the origin group, you configure CloudFront to failover to the second origin for GET, HEAD, and OPTIONS HTTP methods when the primary origin returns specific status codes that you configure.
 - CloudFront is optimized for both dynamic and static content, providing extensive flexibility for optimizing cache behavior, coupled with network-layer optimizations for latency and throughput.
- **Using HTTPS with CloudFront**
 - You can choose HTTPS settings both for communication between viewers and CloudFront, and between CloudFront and your origin.
 - If you want your viewers to use HTTPS and to use alternate domain names for your files, you need to choose one of the following options for how CloudFront serves HTTPS requests:
 - Use a dedicated IP address in each edge location
 - Use Server Name Indication (SNI)
- **Monitoring**
 - The billing report is a high-level view of all of the activity for the AWS services that you're using, including CloudFront.
 - The usage report is a summary of activity for a service such as CloudFront, aggregated by hour, day, or month. It also includes usage charts that provide a graphical representation of your CloudFront usage.
 - CloudFront console includes a variety of reports based on the data in CloudFront access logs:
 - CloudFront Cache Statistics Reports
 - CloudFront Popular Objects Report
 - CloudFront Top Referrers Report
 - CloudFront Usage Reports
 - CloudFront Viewers Reports
 - You can use AWS Config to record configuration changes for CloudFront distribution settings changes.
 - CloudFront integrates with Amazon CloudWatch metrics so that you can monitor your website or application.
 - Capture API requests with AWS CloudTrail. CloudFront is a global service. To view CloudFront requests in CloudTrail logs, you must update an existing trail to include global services.



- **Security**

- CloudFront, AWS Shield, AWS WAF, and Route 53 work seamlessly together to create a flexible, layered security perimeter against multiple types of attacks including network and application layer DDoS attacks.
- You can deliver your content, APIs or applications via SSL/TLS, and advanced SSL features are enabled automatically.
- Through geo-restriction capability, you can prevent users in specific geographic locations from accessing content that you're distributing through CloudFront.
- With the **Origin Access Control** feature, you can restrict access to an S3 bucket to only be accessible from CloudFront.
- **Field-Level Encryption** is a feature of CloudFront that allows you to securely upload user-submitted data such as credit card numbers to your origin servers.

- **Pricing**

- Charge for storage in an S3 bucket.
- Charge for serving objects from edge locations.
- Charge for submitting data to your origin.
 - Data Transfer Out
 - HTTP/HTTPS Requests
 - Invalidation Requests,
 - Dedicated IP Custom SSL certificates associated with a CloudFront distribution.
- You also incur a surcharge for HTTPS requests, and an additional surcharge for requests that also have field-level encryption enabled.

- **Compliance**

- CloudFront has been validated as being compliant with Payment Card Industry (PCI) Data Security Standard (DSS).
- CloudFront is a HIPAA eligible service.
- CloudFront is compliant with SOC measures.

References:

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide>

<https://aws.amazon.com/cloudfront/features/>

<https://aws.amazon.com/cloudfront/pricing/>

<https://aws.amazon.com/cloudfront/faqs/>



AWS Direct Connect

- Using Direct Connect, data can now be delivered through a private network connection between AWS and your datacenter or corporate network.
- Direct Connect links your internal network to a Direct Connect location over a standard Ethernet fiber-optic cable. One end of the cable is connected to your router, the other to a Direct Connect router. With this connection, you can create *virtual interfaces* directly to public AWS services or to Amazon VPC.
- 1 Gbps, 10 Gbps, and 100 Gbps connections are available.
- Supports hosted connection capacities of 1, 2, 5 and 10 Gbps. 1, 2, 5 and 10 Gbps hosted connections will provide customers with higher capacities that were previously only available via dedicated connections.
- Amazon Direct Connect also supports AWS Transit Gateway, aside from configuring Site-to-Site VPN connections. With this feature, customers can connect thousands of Amazon VPCs in multiple AWS Regions to their on-premises networks using 1/2/5/10 Gbps AWS Direct Connect connections.
- **Beneficial Use Cases**
 - When transferring large data sets.
 - When developing and using applications that use real-time data feeds.
 - When building hybrid environments that satisfy regulatory requirements requiring the use of private connectivity.
- **Setting Up Methods**

Port speed	Method
1 Gbps or higher	Connect directly to an AWS device from your router at an AWS Direct Connect location.
1 Gbps or higher	Work with a partner in the AWS Partner Network or a network provider to connect a router from your data center, office, or colocation environment to an AWS Direct Connect location. The network provider does not have to be a member of the APN to connect you.



Less than 1 Gbps	Work with a partner in the AWS Partner Network who can create a hosted connection for you. Sign up for AWS and then follow the instructions to accept your hosted connection.
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- **Components**

- **Connections** - Create a connection in an AWS Direct Connect location to establish a network connection from your premises to an AWS Region. From Direct Connect you can connect to all AZs within the region.
- **Virtual interfaces** - Create a virtual interface to enable access to AWS services. A public virtual interface enables access to public services, such as S3. A private virtual interface enables access to your VPC.
- To access public resources in a remote Region, you must set up a public virtual interface and establish a **Border Gateway Protocol** session.
- You can create a **Direct Connect gateway** in any public Region. Use it to connect your Direct Connect connection over a private virtual interface to VPCs in your account that are located in different Regions.
- To provide for failover, request and configure two dedicated connections to AWS. These connections can terminate on one or two routers in your network. There are different configuration choices available:
 - **Active/Active (BGP multipath)** - This is the default configuration, where both connections are active. If one connection becomes unavailable, all traffic is routed through the other connection.
 - **Active/Passive (failover)** - One connection is handling traffic, and the other is on standby. If the active connection becomes unavailable, all traffic is routed through the passive connection.
- **Autonomous System numbers (ASN)** are used to identify networks that present a clearly defined external routing policy to the Internet.
- **Cross Connects**
 - After you have downloaded your Letter of Authorization and Connecting Facility Assignment (LOA-CFA), you must complete your cross-network connection, also known as a **cross connect**.
 - If you already have equipment located in a Direct Connect location, contact the appropriate provider to complete the cross connect.
 - If you do not already have equipment located in a Direct Connect location, you can work with one of the partners in the AWS Partner Network to help you to connect to an AWS Direct Connect location.
- **Virtual Interfaces**
 - You must create a virtual interface to begin using your Direct Connect connection.
 - You can configure multiple virtual interfaces on a single AWS Direct Connect connection.
 - For private virtual interfaces, you need **one private virtual interface for each VPC** to connect to from the AWS Direct Connect connection, or you can use a **AWS Direct Connect gateway**.



- Prerequisites
 - Connection: The Direct Connect connection or link aggregation group for which you are creating the virtual interface.
 - Virtual interface name: A name for the virtual interface.
 - Virtual interface owner
 - (Private virtual interface only) Connection to
 - VLAN: A unique virtual local area network tag that's not already in use on your connection.
 - Address family: Whether the BGP peering session will be over IPv4 or IPv6.
 - Peer IP addresses: A virtual interface can support a BGP peering session for IPv4, IPv6, or one of each (dual-stack). You cannot create multiple BGP sessions for the same IP addressing family on the same virtual interface
 - BGP information: A public or private Border Gateway Protocol Autonomous System Number for your side of the BGP session, and an MD5 BGP authentication key.
 - (Public virtual interface only) Prefixes you want to advertise: Public IPv4 routes or IPv6 routes to advertise over BGP. You must advertise at least one prefix using BGP.
- The maximum transmission unit (MTU) of a network connection is the size, in bytes, of the largest permissible packet that can be passed over the connection. The MTU of a virtual private interface can be either 1500 or 9001 (jumbo frames). The MTU of a transit virtual interface for VPC Transit Gateways associated with Direct Connect gateways can be either 1500 or 8500 (jumbo frames). A public virtual interface doesn't support jumbo frames.
- Jumbo frames are supported on virtual private interfaces attached to a virtual private gateway or a Direct Connect gateway. Jumbo frames apply only to propagated routes from Direct Connect.
- **Link Aggregation Groups (LAG)**
 - A logical interface that uses the Link Aggregation Control Protocol to aggregate multiple connections at a single Direct Connect endpoint, allowing you to treat them as a single, managed connection.
 - All connections in the LAG must use the same bandwidth.
 - You can have a maximum of four connections in a LAG. Each connection in the LAG counts towards your overall connection limit for the Region.
 - All connections in the LAG must terminate at the same Direct Connect endpoint.
 - Can aggregate up to 4 Direct Connect ports into a single connection using LAG.
 - All connections in a LAG operate in Active/Active mode.
 - It will only be available for dedicated 1G and 10G connections.
- **Direct Connect Gateways**
 - Use a Direct Connect gateway to connect your Direct Connect connection over a private virtual interface to one or more VPCs in your account that are located in the same or different Regions.
 - It is a globally available resource.



- Direct Connect gateway also enables you to connect between your on-premises networks and Amazon Virtual Private Cloud (Amazon VPC) in any commercial AWS Region except in China regions.
- Prior to multi-account support, you could only associate Amazon VPCs with a Direct Connect gateway in the same AWS account. With the launch of multi-account support for Direct Connect gateway, you can associate up to 10 Amazon VPCs from multiple accounts with a Direct Connect gateway. The VPCs must be owned by AWS Accounts that belong to the same AWS payer account ID.
- **Security**
 - Use IAM for controlling access.
- **Monitoring**
 - You can optionally assign tags to your Direct Connect resources to categorize or manage them. A tag consists of a key and an optional value, both of which you define.
 - CloudTrail captures all API calls for AWS Direct Connect as events.
 - Set up CloudWatch alarms to monitor metrics.
- **Pricing**
 - You pay only for the network ports you use and the data you transfer over the connection.
 - Pricing is per port-hour consumed for each port type. Data transfer out over AWS Direct Connect is charged per GB. Data transfer IN is \$0.00 per GB in all locations.

References:

<https://docs.aws.amazon.com/directconnect/latest/UserGuide>

<https://aws.amazon.com/directconnect/features/>

<https://aws.amazon.com/directconnect/pricing/>

<https://aws.amazon.com/directconnect/faqs/>

AWS Transit Gateway

- A networking service that uses a hub and spoke model to enable customers to connect their on-premises data centers and their Amazon Virtual Private Clouds (VPCs) to a single gateway.
- With this service, customers only have to create and manage a single connection from the central gateway into each on-premises data center, remote office, or VPC across your network.
- If a new VPC is created, it is automatically connected to the Transit Gateway and will also be available to every other network that is also connected to the Transit Gateway.

Features:

- **Inter-region peering**
 - Transit Gateway leverages the AWS global network to allow customers to route traffic across AWS Regions.



- Inter-region peering provides an easy and cost-effective way to replicate data for geographic redundancy or to share resources between AWS Regions.
- **Multicast**
 - Enables customers to have fine-grain control on who can consume and produce multicast traffic.
 - It allows you to easily create and manage multicast groups in the cloud instead of the time-consuming task of deploying and managing legacy hardware on-premises.
 - This multicast solution is also scalable so the customers can simultaneously distribute a stream of content to multiple subscribers.
- **Automated Provisioning**
 - Customers can automatically identify the Site-to-Site VPN connections and the on-premises resources with which they are associated using AWS Transit Gateway.
 - Using the Transit Gateway Network Manager, you can also manually define your on-premises network.

Reference:

<https://aws.amazon.com/transit-gateway/>

AWS Organizations

- It offers policy-based management for multiple AWS accounts.

Features

- With Organizations, you can create groups of accounts and then apply policies to those groups.
- Organizations provides you a policy framework for multiple AWS accounts. You can apply policies to a group of accounts or all the accounts in your organization.
- AWS Organizations enables you to set up a single payment method for all the AWS accounts in your organization through **consolidated billing**. With consolidated billing, you can see a combined view of charges incurred by all your accounts, as well as take advantage of pricing benefits from aggregated usage, such as volume discounts for EC2 and S3.
- AWS Organizations, like many other AWS services, is **eventually consistent**. It achieves high availability by replicating data across multiple servers in AWS data centers within its region.

Administrative Actions in Organizations

- Create an AWS account and add it to your organization, or add an existing AWS account to your organization.
- Organize your AWS accounts into groups called *organizational units* (OUs).
- Organize your OUs into a hierarchy that reflects your company's structure.
- Centrally manage and attach policies to the entire organization, OUs, or individual AWS accounts.



Concepts

- An **organization** is a collection of AWS accounts that you can organize into a hierarchy and manage centrally.
- A **management account** is the AWS account you use to create your organization. You cannot change which account in your organization is the management account.
 - From the management account, you can create other accounts in your organization, invite and manage invitations for other accounts to join your organization, and remove accounts from your organization.
 - You can also attach policies to entities such as administrative roots, organizational units (OUs), or accounts within your organization.
 - The management account has the role of a payer account and is responsible for paying all charges accrued by the accounts in its organization.
- A **member account** is an AWS account, other than the management account, that is part of an organization. A member account can belong to only one organization at a time. The management account has the responsibilities of a payer account and is responsible for paying all charges that are accrued by the member accounts.
- An **administrative root** is the starting point for organizing your AWS accounts. The administrative root is the top-most container in your organization's hierarchy. Under this root, you can create OUs to logically group your accounts and organize these OUs into a hierarchy that best matches your business needs.
- An **organizational unit (OU)** is a group of AWS accounts within an organization. An OU can also contain other OUs enabling you to create a hierarchy.
- A **policy** is a "document" with one or more statements that define the controls that you want to apply to a group of AWS accounts.
 - **Service control policy (SCP)** is a policy that specifies the services and actions that users and roles can use in the accounts that the SCP affects. SCPs are similar to IAM permission policies except that they don't grant any permissions. Instead, SCPs are *filters* that allow only the specified services and actions to be used in affected accounts.
- AWS Organizations has two available feature sets:
 - All organizations support **consolidated billing**, which provides basic management tools that you can use to centrally manage the accounts in your organization.
 - If you enable **all features**, you continue to get all the consolidated billing features plus a set of advanced features such as service control policies.
- You can remove an AWS account from an organization and make it into a standalone account.
- Organization Hierarchy
 - Including root and AWS accounts created in the lowest OUs, your hierarchy can be five levels deep.
 - Policies inherited through hierarchical connections in an organization.
 - Policies can be assigned at different points in the hierarchy.



Pricing

- This service is free.

References:

<https://docs.aws.amazon.com/organizations/latest/userguide/>

<https://aws.amazon.com/organizations/features/>

<https://aws.amazon.com/organizations/faqs/>

AWS Control Tower

- A service for configuring and managing a multi-account AWS environment.

Concepts

- Landing zone
 - A multi-account environment that is well-architected and adheres to security and compliance best practices.
 - Each organization can have one landing zone.
 - A container that holds the following:
 - Organizational Units (OUs)
 - Accounts
 - Users
 - Other Resources
 - Structure of a landing zone:
 - Root – parent that contains all OUs.
 - Security OU – contains the shared accounts.
 - Sandbox OU – contain the registered accounts used by your users to carry out their AWS workloads.
 - IAM Identity Center directory – scope of permissions of each user.
 - IAM Identity Center users – identities that your users can use to perform AWS workloads.
- Guardrails
 - A high-level rule or policy that governs your AWS environment.
 - Applies to both OU and AWS accounts within the OU.
 - Guardrails are classified based on their behavior and guidance.
 - Behavior
 - Preventive
 - Prohibits actions that result in policy violations.
 - Implemented using AWS Organizations SCPs.
 - Status is either enforced or not enabled.
 - Detective
 - Detects noncompliance resources and provides alerts through the dashboard.



- Implemented using AWS Config rules.
 - Status is either clear, in violation, or not enabled.
- Guidance
 - Mandatory – always enforced.
 - Strongly recommended – enforce best practices.
 - Elective – track actions that are commonly restricted.
- By default, mandatory guardrails are applied to top-level OUs.
- The exception for guardrails is only for root or management accounts.
- Account Factory
 - Automates provisioning of new accounts.
 - It also helps you standardize the provisioning of new accounts by using pre-approved account configurations.
 - Shared accounts:
 - Management account – used for billing, provisioning of accounts, and managing OUs and guardrails.
 - Log Archive account – a repository for logs of API activities and resource configurations.
 - Audit account – a restricted account for security and compliance teams.
 - Member accounts are the accounts used by your users to perform AWS workloads.
 - You can also provision accounts using AWS Control Tower Account Factory for Terraform.
- Dashboard
 - Offers continuous oversight to the following:
 - Accounts across your enterprise.
 - Guardrails enabled for policy enforcement.
 - Guardrails enabled for continuous detection of policy non-conformance.
 - Non-compliant resources organized by accounts and OUs.

AWS Control Tower Networking Features

- AWS automatically creates an AWS-default VPC in every Region, even those not governed by AWS Control Tower, as part of the account creation process.
- The default VPC is not the same as the VPC created by AWS Control Tower for a provisioned account.
- You also have the option to remove AWS default VPCs in non-governed Regions.
- Each AWS Control Tower VPC has three Availability Zones.
- By default, an AZ has one public and two private subnets.
- Supports VPC-to-VPC peering for multiple VPCs.
- Region deny guardrail
 - Applies to a landing zone.
 - Blocks API calls to services in non-governed Regions.



- IAM users can still connect to an AWS default VPC in a Region where AWS Control Tower is not supported.
- If a guardrail is enabled, you will be unable to deploy resources in the denied Regions.

AWS Control Tower Monitoring

- A log archive account is dedicated to collecting all logs centrally.
- You can use AWS CloudTrail to capture the actions or events of AWS Control Tower.
- With CloudWatch Logs and CloudWatch Logs Insights, you can view and query AWS Control Tower lifecycle events.
- A lifecycle event is only recorded after a series of actions has been completed.
- The event log for each lifecycle event indicates whether the originating Control Tower action was successful or unsuccessful.
- Each lifecycle event is automatically recorded as a non-API AWS service event by AWS CloudTrail.
- Each lifecycle event is sent to Amazon EventBridge.

AWS Control Tower Pricing

- You are charged for AWS services that are configured to set up your landing zone and mandatory guardrails.
- You are charged by AWS Config for running ephemeral workloads as it records configuration changes related to the creation and deletion of temporary resources.

AWS CloudFormation

- A service that gives developers and businesses an easy way to create a collection of related AWS resources and provision them in an orderly and predictable fashion.

Features

- CloudFormation allows you to model your entire infrastructure in a text file called a **template**. You can use JSON or YAML to describe what AWS resources you want to create and configure. If you want to design visually, you can use AWS Infrastructure Composer.
- CloudFormation automates the provisioning and updating of your infrastructure in a safe and controlled manner. You can use **Rollback Triggers** to specify the CloudWatch alarm that CloudFormation should monitor during the stack creation and update process. If any of the alarms are breached, CloudFormation rolls back the entire stack operation to a previously deployed state.
- **CloudFormation Change Sets** allow you to preview how proposed changes to a stack might impact your running resources.
- **AWS StackSets** lets you provision a common set of AWS resources across multiple accounts and regions with a single CloudFormation template. StackSets takes care of automatically and safely provisioning, updating, or deleting stacks in multiple accounts and across multiple regions.



- CloudFormation enables you to build custom extensions to your stack template using AWS Lambda.

CloudFormation vs Elastic Beanstalk

- Elastic Beanstalk provides an **environment** to easily **deploy and run** applications in the cloud.
- CloudFormation is a convenient **provisioning mechanism** for a broad range of AWS resources.

Concepts

- **Templates**
 - A JSON or YAML formatted text file.
 - CloudFormation uses these templates as blueprints for building your AWS resources.
- **Stacks**
 - Manage related resources as a single unit.
 - All the resources in a stack are defined by the stack's CloudFormation template.
- **Change Sets**
 - Before updating your stack and making changes to your resources, you can generate a change set, which is a summary of your proposed changes.
 - Change sets allow you to see how your changes might impact your running resources, especially for critical resources, before implementing them.
- With AWS CloudFormation and AWS CodePipeline, you can use continuous delivery to automatically build and test changes to your CloudFormation templates before promoting them to production stacks.
- *CloudFormation artifacts* can include a stack template file, a template configuration file, or both. AWS CodePipeline uses these artifacts to work with CloudFormation stacks and change sets.
 - **Stack Template File** - defines the resources that CloudFormation provisions and configures. You can use YAML or JSON-formatted templates.
 - **Template Configuration File** - a JSON-formatted text file that can specify template parameter values, a stack policy, and tags. Use these configuration files to specify parameter values or a stack policy for a stack.
- Through the AWS PrivateLink, you can use CloudFormation APIs inside of your Amazon VPC and route data between your VPC and CloudFormation entirely within the AWS network.

Stacks

- If a resource cannot be created, CloudFormation rolls the stack back and automatically deletes any resources that were created. If a resource cannot be deleted, any remaining resources are retained until the stack can be successfully deleted.
- Stack update methods
 - Direct update
 - Creating and executing change sets



- **Drift detection** enables you to detect whether a stack's actual configuration differs, or has drifted, from its expected configuration. Use CloudFormation to detect drift on an entire stack, or on individual resources within the stack.
 - A resource is considered to have drifted if any of its actual property values differ from the expected property values.
 - A stack is considered to have drifted if one or more of its resources have drifted.
- To share information between stacks, export a stack's output values. Other stacks that are in the same AWS account and region can import the exported values.
- You can nest stacks.

Templates

- Templates include several major sections. The Resources section is the only required section.
- **AWS Infrastructure Composer** is a graphic tool for creating, viewing, and modifying CloudFormation templates. You can diagram your template resources using a drag-and-drop interface, and then edit their details using the integrated JSON and YAML editor.
- Custom resources enable you to write custom provisioning logic in templates that CloudFormation runs anytime you create, update (if you changed the custom resource), or delete stacks.
- Template macros enable you to perform custom processing on templates, from simple actions like find-and-replace operations to extensive transformations of entire templates.

StackSets

- CloudFormation StackSets allow you to roll out CloudFormation stacks over multiple AWS accounts and in multiple Regions with just a couple of clicks. StackSets is commonly used together with AWS Organizations to centrally deploy and manage services in different accounts.
- Administrator and target accounts - An *administrator account* is the AWS account in which you create stack sets. A stack set is managed by signing in to the AWS administrator account in which it was created. A *target account* is the account into which you create, update, or delete one or more stacks in your stack set.
- In addition to the organization's management account, you can delegate other administrator accounts in your AWS Organization that can create and manage stack sets with service-managed permissions for the organization.
- Stack sets - A *stack set* lets you create stacks in AWS accounts across regions by using a single CloudFormation template. All the resources included in each stack are defined by the stack set's CloudFormation template. A stack set is a regional resource.
- Stack instances - A *stack instance* is a reference to a stack in a target account within a region. A stack instance can exist without a stack; for example, if the stack could not be created for some reason, the stack instance shows the reason for stack creation failure. A stack instance can be associated with only one stack set.
- Stack set operations - Create stack set, update stack set, delete stacks, and delete stack set.



- Tags - You can add tags during stack set creation and update operations by specifying key and value pairs.

Monitoring

- CloudFormation is integrated with AWS CloudTrail, a service that provides a record of actions taken by a user, role, or an AWS service in CloudFormation. CloudTrail captures all API calls for CloudFormation as events, including calls from the CloudFormation console and from code calls to the CloudFormation APIs.

Security

- You can use IAM with CloudFormation to control what users can do with AWS CloudFormation, such as whether they can view stack templates, create stacks, or delete stacks.
- A *service role* is an IAM role that allows CloudFormation to make calls to resources in a stack on your behalf. You can specify an IAM role that allows CloudFormation to create, update, or delete your stack resources.
- You can improve the security posture of your VPC by configuring CloudFormation to use an interface VPC endpoint.

Pricing

- No additional charge for CloudFormation. You pay for AWS resources created using CloudFormation in the same manner as if you created them manually.

References:

<https://docs.aws.amazon.com/AWSCloudFormation/latest/UserGuide/>

<https://aws.amazon.com/cloudformation/features/>

<https://aws.amazon.com/cloudformation/pricing/>

<https://aws.amazon.com/cloudformation/faqs/>

AWS Service Catalog

- Allows you to create, manage, and distribute catalogs of approved products to end-users, who can then access the products they need in a personalized portal.
- Administrators can control which users have access to each product to enforce compliance with organizational business policies. Administrators can also set up adopted roles so that end users only require IAM access to AWS Service Catalog in order to deploy approved resources.
- This is a regional service.

Features

- Standardization of assets



- Self-service discovery and launch
- Fine-grain access control
- Extensibility and version control

Concepts

- Users
 - Catalog administrators – Manage a catalog of products, organizing them into portfolios and granting access to end users. Catalog administrators prepare AWS CloudFormation templates, configure constraints, and manage IAM roles that are assigned to products to provide for advanced resource management.
 - End users – Use AWS Service Catalog to launch products to which they have been granted access.
- Products
 - Can comprise one or more AWS resources, such as EC2 instances, storage volumes, databases, monitoring configurations, and networking components, or packaged AWS Marketplace products.
 - You create your products by importing AWS CloudFormation templates. The templates define the AWS resources required for the product, the relationships between resources, and the parameters for launching the product to configure security groups, create key pairs, and perform other customizations.
 - You can see the products that you are using and their health state in the AWS Service Catalog console.
- Portfolio
 - A collection of products, together with configuration information. Portfolios help manage product configuration, determine who can use specific products and how they can use them.
 - When you add a new version of a product to a portfolio, that version is automatically available to all current users of that portfolio.
 - You can also share your portfolios with other AWS accounts and allow the administrator of those accounts to distribute your portfolios with additional constraints.
 - When you add tags to your portfolio, the tags are applied to all instances of resources provisioned from products in the portfolio.
- Versioning
 - Service Catalog allows you to manage multiple versions of the products in your catalog.
 - A version can have one of three statuses:
 - Active - An active version appears in the version list and allows users to launch it.
 - Inactive - An inactive version is hidden from the version list. Existing provisioned products launched from this version will not be affected.
 - Deleted - If a version is deleted, it is removed from the version list. Deleting a version can't be undone.
- Access control



- You apply AWS IAM permissions to control who can view and modify your products and portfolios.
- By assigning an IAM role to each product, you can avoid giving users permissions to perform unapproved operations, and enable them to provision resources using the catalog.
- Constraints
 - You use constraints to apply limits to products for governance or cost control.
 - Types of constraints:
 - Template constraints restrict the configuration parameters that are available for the user when launching the product. Template constraints allow you to reuse generic AWS CloudFormation templates for products and apply restrictions to the templates on a per-product or per-portfolio basis.
 - Launch constraints allow you to specify a role for a product in a portfolio. This role is used to provision the resources at launch, so you can restrict user permissions without impacting users' ability to provision products from the catalog.
 - Notification constraints specify an Amazon SNS topic to receive notifications about stack events.
 - Tag update constraints allow administrators to allow or disallow end users to update tags on resources associated with an AWS Service Catalog provisioned product.
- Stack
 - Every AWS Service Catalog product is launched as an AWS CloudFormation stack.
 - You can use CloudFormation StackSets to launch Service Catalog products across multiple regions and accounts. You can specify the order in which products deploy sequentially within regions. Across accounts, products are deployed in parallel.

Security

- Service Catalog uses Amazon S3 buckets and Amazon DynamoDB databases that are encrypted at rest using Amazon-managed keys.
- Service Catalog uses TLS and client-side encryption of information in transit between the caller and AWS.
- Service Catalog integrates with AWS CloudTrail and Amazon SNS.

Pricing

- The AWS Service Catalog free tier includes 1,000 API calls per month.
- You are charged based on the number of API calls made to Service Catalog beyond the free tier.

References:

<https://aws.amazon.com/servicecatalog/>

<https://docs.aws.amazon.com/servicecatalog/latest/adminguide/introduction.html>

<https://docs.aws.amazon.com/servicecatalog/latest/userguide/end-user-console.html>



<https://aws.amazon.com/servicecatalog/pricing/>

<https://aws.amazon.com/servicecatalog/faqs/>

AWS Systems Manager

- Allows you to centralize operational data from multiple AWS services and automate tasks across your AWS resources.

Features

- Create logical groups of resources such as applications, different layers of an application stack, or production versus development environments.
- You can select a resource group and view its recent API activity, resource configuration changes, related notifications, operational alerts, software inventory, and patch compliance status.
- Collects information about your instances and the software installed on them.
- Allows you to safely automate common and repetitive IT operations and management tasks across AWS resources.
- Provides a browser-based interactive shell and CLI for managing Windows and Linux EC2 instances, without the need to open inbound ports, manage SSH keys, or use bastion hosts. Administrators can grant and revoke access to instances through a central location by using IAM policies.
- Helps ensure that your software is up-to-date and meets your compliance policies.
- Lets you schedule windows of time to run administrative and maintenance tasks across your instances.

SSM Agent is the tool that processes Systems Manager requests and configures your machine as specified in the request. SSM Agent must be installed on each instance you want to use with Systems Manager. On newer AMIs and instance types, SSM Agent is installed by default. On older versions, you must install it manually.

Capabilities

- **Automation**
 - Allows you to safely automate common and repetitive IT operations and management tasks across AWS resources
 - A **step** is defined as an initiated action performed in the Automation execution on a per-target basis. You can execute the entire Systems Manager automation document in one action or choose to execute one step at a time.
 - Concepts
 - **Automation document** - defines the Automation workflow.
 - **Automation action** - the Automation workflow includes one or more steps. Each step is associated with a particular action or plugin. The action determines the inputs, behavior, and outputs of the step.
 - **Automation queue** - if you attempt to run more than 25 Automations simultaneously, Systems Manager adds the additional executions to a queue and

displays a status of *Pending*. When an Automation reaches a terminal state, the first execution in the queue starts.

- You can schedule Systems Manager automation document execution.

- **Resource Groups**

- A collection of AWS resources that are all in the same AWS region, and that match criteria provided in a query.
- Use Systems Manager tools such as *Automation* to simplify management tasks on your groups of resources. You can also use groups as the basis for viewing monitoring and configuration *insights* in Systems Manager.

- **Built-in Insights**

- Show detailed information about a single, selected resource group.
- Includes recent API calls through CloudTrail, recent configuration changes through Config, Instance software inventory listings, instance patch compliance views, and instance configuration compliance views.

- **Systems Manager Activation**

- Enable hybrid and cross-cloud management. You can register any server, whether physical or virtual to be managed by Systems Manager.

- **Inventory Manager**

- Automates the process of collecting software inventory from managed instances.
- You specify the type of metadata to collect, the instances from where the metadata should be collected, and a schedule for metadata collection.

- **Configuration Compliance**

- Scans your fleet of managed instances for patch compliance and configuration inconsistencies.
- View compliance history and change tracking for Patch Manager patching data and State Manager associations by using AWS Config.
- Customize Systems Manager Compliance to create your own compliance types.

- **Run Command**

- Remotely and securely manage the configuration of your managed instances at scale.
- **Managed Instances** - any EC2 instance or on-premises server or virtual machine in your hybrid environment that is configured for Systems Manager.

- **Session Manager**

- Manage your EC2 instances through an interactive one-click browser-based shell or through the AWS CLI.
- Makes it easy to comply with corporate policies that require controlled access to instances, strict security practices, and fully auditable logs with instance access details, while still providing end users with simple one-click cross-platform access to your Amazon EC2 instances.



- You can use AWS Systems Manager Session Manager to tunnel SSH (Secure Shell) and SCP (Secure Copy) traffic between a client and a server.
- **Distributor**
 - Lets you package your own software or find AWS-provided agent software packages to install on Systems Manager managed instances.
 - After you create a package in Distributor, which creates an Systems Manager document, you can install the package in one of the following ways.
 - One time by using Systems Manager Run Command.
 - On a schedule by using Systems Manager State Manager.
- **Patch Manager**
 - Automate the process of patching your managed instances.
 - Enables you to scan instances for missing patches and apply missing patches individually or to large groups of instances by using EC2 instance tags.
 - For security patches, Patch Manager uses *patch baselines* that include rules for auto-approving patches within days of their release, as well as a list of approved and rejected patches.
 - You can use AWS Systems Manager Patch Manager to select and apply Microsoft application patches automatically across your Amazon EC2 or on-premises instances.
 - AWS Systems Manager Patch Manager includes common vulnerability identifiers (CVE ID). CVE IDs can help you identify security vulnerabilities within your fleet and recommend patches.
 - You can configure actions to be performed on a managed instance before and after installing patches.
- **Incident Manager**
 - Provides an incident management console to manage and monitor all incidents relating to the AWS resources that your applications are using.
 - Primarily used to mitigate and recover from production incidents affecting their applications that are hosted in AWS.
 - Expedites incident resolution by notifying responders of impact, highlighting relevant troubleshooting data, and providing collaboration tools to return normal operations quickly.
 - Automates response plans and allows responder team escalation.
- **Compliance**
 - Automatically scans your fleet of managed nodes in AWS for patch compliance and configuration inconsistencies.
 - Collects and aggregates data from multiple AWS accounts and AWS Regions then drill down into specific resources that aren't compliant.
 - Displays compliance data about Patch Manager patching and State Manager associations.
 - Allows you to create your own compliance types based on your technical requirements.



- **Fleet Manager**

- A unified user interface (UI) experience that helps you remotely manage your nodes/servers in AWS which allows you to view the health and performance status of your entire fleet from a single UI console.
- Gathers data from individual devices, external servers and Amazon EC2 instances to perform common troubleshooting and management tasks straight from the console without manually connecting to the resource.
- Enables you to view the directory and file contents of your nodes/instances, Windows registry management, operating system user management et cetera.

- **State Manager**

- A service that automates the process of keeping your EC2 and hybrid infrastructure in a state that you define.
- A *State Manager association* is a configuration that is assigned to your managed instances. The configuration defines the state that you want to maintain on your instances. The association also specifies actions to take when applying the configuration.

- **Parameter Store**

- Provides secure, hierarchical storage for configuration data and secrets management.
- You can store values as plain text or encrypted data with *SecureString*.
- Parameters work with Systems Manager capabilities such as Run Command, State Manager, and Automation.

- **OpsCenter**

- OpsCenter helps you view, investigate, and resolve operational issues related to your environment from a central location.
- OpsCenter complements existing case management systems by enabling integrations via Amazon Simple Notification Service (SNS) and public AWS SDKs. By aggregating information from AWS Config, AWS CloudTrail logs, resource descriptions, and Amazon EventBridge (Amazon CloudWatch Events), OpsCenter helps you reduce the mean time to resolution (MTTR) of incidents, alarms, and operational tasks.

- **Change Manager**

- An enterprise change management framework for requesting, approving, implementing, and reporting on operational changes to your application configuration and infrastructure.
- From a single delegated administrator account, if you use AWS Organizations, you can manage changes across multiple AWS accounts and across AWS Regions. Alternatively, using a local account, you can manage changes for a single AWS account.
- Can be used for both AWS and on-premises resources.



- For each change template, you can add up to five levels of approvers. When it's time to implement an approved change, Change Manager runs the Automation runbook that is specified in the associated change request.

- **Maintenance Window**

- Set up recurring schedules for managed instances to execute administrative tasks like installing patches and updates without interrupting business-critical operations.
- Supports running four types of tasks:
 - Systems Manager Run Command commands
 - Systems Manager Automation workflows
 - AWS Lambda functions
 - AWS Step Functions tasks

- **Systems Manager Document (SSM)**

- Defines the actions that Systems Manager performs.
- Types of SSM Documents

Type	Use with	Details
Command document	Run Command, State Manager	Run Command uses command documents to execute commands. State Manager uses command documents to apply a configuration. These actions can be run on one or more targets at any point during the lifecycle of an instance.
Policy document	State Manager	Policy documents enforce a policy on your targets. If the policy document is removed, the policy action no longer happens.
Automation document	Automation	Use automation documents when performing common maintenance and deployment tasks such as creating or updating an AMI.
Package document	Distributor	In Distributor, a package is represented by a Systems Manager document. A package document includes attached ZIP archive files that contain software or assets to install on managed instances. Creating a package in Distributor creates the package document.



- Can be in JSON or YAML.
- You can create and save different versions of documents. You can then specify a default version for each document.
- If you want to customize the steps and actions in a document, you can create your own.
- You can tag your documents to help you quickly identify one or more documents based on the tags you've assigned to them.

Monitoring

- SSM Agent writes information about executions, scheduled actions, errors, and health statuses to log files on each instance. For more efficient instance monitoring, you can configure either SSM Agent itself or the CloudWatch Agent to send this log data to CloudWatch Logs.
- Using CloudWatch Logs, you can monitor log data in real-time, search and filter log data by creating one or more metric filters, and archive and retrieve historical data when you need it.
- Log System Manager API calls with CloudTrail.

Security

- Systems Managers is linked directly to IAM for access controls.

Pricing

- For your own packages, you pay only for what you use. Upon transferring a package into Distributor, you will be charged based on the size and duration of storage for that package, the number of Get and Describe API calls made, and the amount of out-of-Region and on-premises data transfer out of Distributor for those packages.
- You are charged based on the number and type of Automation steps.

References:

<https://docs.aws.amazon.com/systems-manager/latest/userguide>

<https://aws.amazon.com/systems-manager/features/>

<https://aws.amazon.com/systems-manager/pricing/>

<https://aws.amazon.com/systems-manager/faq/>



AWS Config

- A fully managed service that provides you with an AWS resource inventory, configuration history, and configuration change notifications to enable security and governance.

Features

- Multi-account, multi-region data aggregation gives you an enterprise-wide view of your **Config rule** compliance status, and you can associate your AWS organization to quickly add your accounts.
- Provides you pre-built rules to evaluate your AWS resource configurations and configuration changes, or create your own custom rules in AWS Lambda that define your internal best practices and guidelines for resource configurations.
- **Config records** details of changes to your AWS resources to provide you with a configuration history, and automatically deliver it to an S3 bucket you specify.
- Receive a notification whenever a resource is created, modified, or deleted.
- Config enables you to record software configuration changes within your EC2 instances and servers running on-premises, as well as servers and Virtual Machines in environments provided by other cloud providers. You gain visibility into:
 - operating system configurations
 - system-level updates
 - installed applications
 - network configuration
- Config can provide you with a **configuration snapshot** - a point-in-time capture of all your resources and their configurations.
- Config discovers, maps, and tracks AWS resource relationships in your account.
Ex. EC2 instances and associated security groups

Concepts

Configuration History

- A collection of the configuration items for a given resource over any time period, containing information such as when the resource was first created, how the resource has been configured over the last month, etc.
- Config automatically delivers a configuration history file for each resource type that is being recorded to an S3 bucket that you specify.
- A configuration history file is sent every six hours for each resource type that Config records.

Configuration item

- A record of the configuration of a resource in your AWS account. Config creates a configuration item whenever it detects a change to a resource type that it is recording.
- The components of a configuration item include metadata, attributes, relationships, current configuration, and related events.



Configuration Recorder

- Stores the configurations of the supported resources in your account as configuration items.
- By default, the configuration recorder records all supported resources in the region where Config is running. You can create a customized configuration recorder that records only the resource types that you specify.
- You can also have Config record supported types of *global resources* which are IAM users, groups, roles, and customer managed policies.

Configuration Snapshot

- A complete picture of the resources that are being recorded and their configurations.
- Stored in an S3 bucket that you specify.

Configuration Stream

- An automatically updated list of all configuration items for the resources that Config is recording.
- Helpful for observing configuration changes as they occur so that you can spot potential problems, generating notifications if certain resources are changed, or updating external systems that need to reflect the configuration of your AWS resources.

Configuration Item

- The configuration of a resource at a given point-in-time. A CI consists of 5 sections:
 - Basic information about the resource that is common across different resource types.
 - Configuration data specific to the resource.
 - Map of relationships with other resources.
 - CloudTrail event IDs that are related to this state.
 - Metadata that helps you identify information about the CI, such as the version of this CI, and when this CI was captured.

Resource Relationship

- Config discovers AWS resources in your account and then creates a map of relationships between AWS resources.

Config rule

- Represents your desired configuration settings for specific AWS resources or for an entire AWS account.
- Provides customizable, predefined rules. If a resource violates a rule, Config flags the resource and the rule as noncompliant, and notifies you through Amazon SNS.
- Evaluates your resources either **in response to configuration changes** or **periodically**.
- Config deletes data older than your specified retention period. The default period is 7 years.
- Multi-Account Multi-Region Data Aggregation
 - An aggregator collects configuration and compliance data from the following:
 - Multiple accounts and multiple regions.
 - Single account and multiple regions.
 - An organization in AWS Organizations and all the accounts in that organization.



Monitoring

- Use Amazon SNS to send you notifications every time a supported AWS resource is created, updated, or otherwise modified as a result of user API activity.
- Use Amazon EventBridge (Amazon CloudWatch Events) to detect and react to changes in the status of AWS Config events.
- Use AWS CloudTrail to capture API calls to Config.

Security

- Use IAM to create individual users for anyone who needs access to Config and grant different permissions to each IAM user.

Compliances

- ISO
- PCI DSS
- HIPAA
- FedRAMP

Pricing

- You are charged based on the number of configuration items recorded and on the number of AWS Config rules evaluations recorded, instead of the number of active rules in your account per region.. You are charged only once for recording the configuration item.

References:

<https://docs.aws.amazon.com/config/latest/developerguide>

<https://aws.amazon.com/config/features/>

<https://aws.amazon.com/config/pricing/>

<https://aws.amazon.com/config/faq/>

Amazon CloudWatch

- Monitoring tool for your AWS resources and applications.
- Display metrics and create alarms that watch the metrics and send notifications or automatically make changes to the resources you are monitoring when a threshold is breached.
- CloudWatch is basically a metrics repository. An AWS service, such as Amazon EC2, puts metrics into the repository and you retrieve statistics based on those metrics. If you put your own custom metrics into the repository, you can retrieve statistics on these metrics as well.
- CloudWatch does not aggregate data across regions. Therefore, metrics are completely separate between regions.
- **CloudWatch Concepts**
 - **Namespaces** - a container for CloudWatch metrics.
 - There is no default namespace.
 - The AWS namespaces use the following naming convention: `AWS/service`.
 - **Metrics** - represents a time-ordered set of data points that are published to CloudWatch.
 - Exists only in the region in which they are created.
 - Cannot be deleted, but they automatically expire after 15 months if no new data is published to them.
 - As new data points come in, data older than 15 months is dropped.
 - Each metric data point must be marked with a *timestamp*. The timestamp can be up to two weeks in the past and up to two hours into the future. If you do not provide a timestamp, CloudWatch creates a timestamp for you based on the time the data point was received.
 - By default, several services provide free metrics for resources. You can also enable **detailed monitoring**, or publish your own application metrics.
 - **Metric math** enables you to query multiple CloudWatch metrics and use math expressions to create new time series based on these metrics.
 - **Important note for EC2 metrics:** CloudWatch does not collect memory utilization and disk space usage metrics right from the get go. You need to install CloudWatch Agent in your instances first to retrieve these metrics.
 - **Dimensions** - a name/value pair that uniquely identifies a metric.
 - You can assign up to 10 dimensions to a metric.
 - Whenever you add a unique dimension to one of your metrics, you are creating a new variation of that metric.
 - **Statistics** - metric data aggregations over specified periods of time.
 - Each statistic has a unit of measure. Metric data points that specify a unit of measure are aggregated separately.
 - You can specify a unit when you create a custom metric. If you do not specify a unit, CloudWatch uses *None* as the unit.



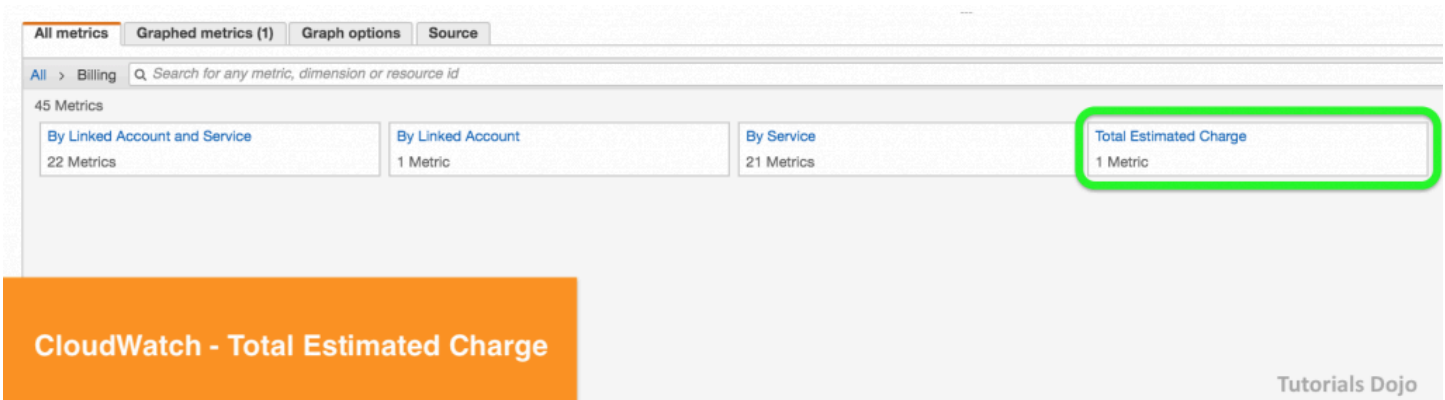
- A *period* is the length of time associated with a specific CloudWatch statistic. The default value is 60 seconds.
- CloudWatch aggregates statistics according to the period length that you specify when retrieving statistics.
- For large datasets, you can insert a pre-aggregated dataset called a *statistic set*.

Statistic	Description
Minimum	The lowest value observed during the specified period. You can use this value to determine low volumes of activity for your application.
Maximum	The highest value observed during the specified period. You can use this value to determine high volumes of activity for your application.
Sum	All values submitted for the matching metric added together. Useful for determining the total volume of a metric.
Average	The value of Sum / SampleCount during the specified period. By comparing this statistic with the Minimum and Maximum, you can determine the full scope of a metric and how close the average use is to the Minimum and Maximum. This comparison helps you to know when to increase or decrease your resources as needed.
SampleCount	The count (number) of data points used for the statistical calculation.
pNN.NN	The value of the specified percentile. You can specify any percentile, using up to two decimal places (for example, p95.45). Percentile statistics are not available for metrics that include any negative values.

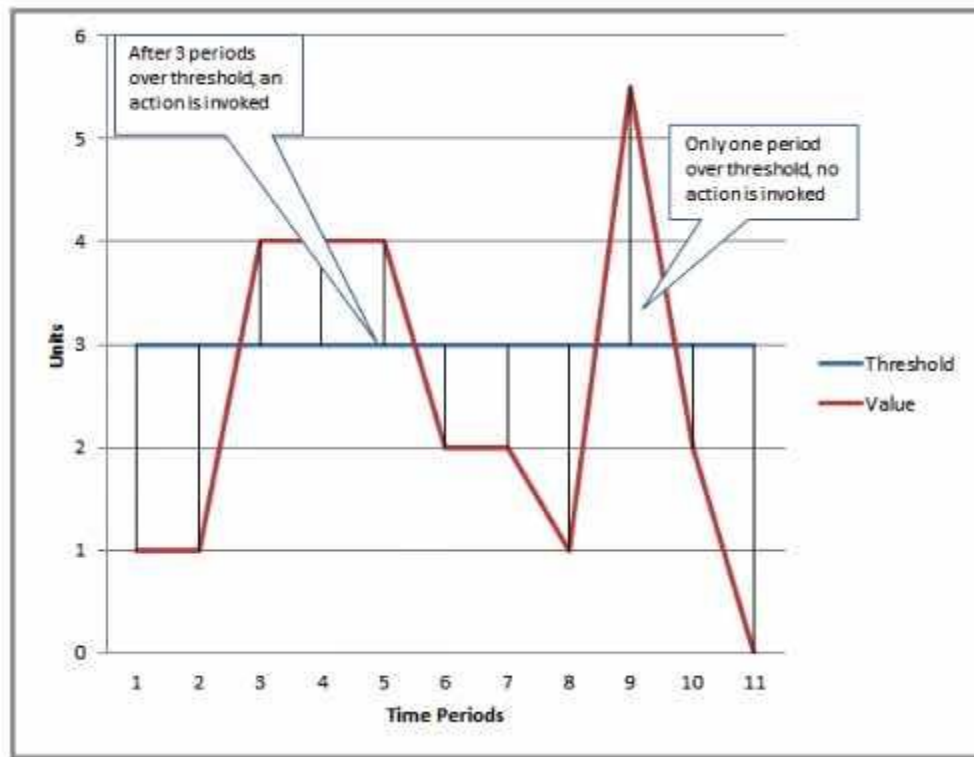
- **Percentiles** - indicates the relative standing of a value in a dataset. Percentiles help you get a better understanding of the distribution of your metric data.
- **Alarms** - watches a single metric over a specified time period, and performs one or more specified actions, based on the value of the metric relative to a threshold over time.



- You can create an alarm for monitoring CPU usage and load balancer latency, for managing instances, and for billing alarms.
- When an alarm is on a dashboard, it turns red when it is in the *ALARM* state.
- Alarms invoke actions for sustained state changes only.
- Alarm States
 - **OK**—The metric or expression is within the defined threshold.
 - **ALARM**—The metric or expression is outside of the defined threshold.
 - **INSUFFICIENT_DATA**—The alarm has just started, the metric is not available, or not enough data is available for the metric to determine the alarm state.
- You can also monitor your estimated AWS charges by using Amazon CloudWatch Alarms. However, take note that you can only track the estimated AWS charges in CloudWatch and not the actual utilization of your resources. Remember that you can only set coverage targets for your reserved EC2 instances in AWS Budgets or Cost Explorer, but not in CloudWatch.



- When you create an alarm, you specify three settings:
 - **Period** is the length of time to evaluate the metric or expression to create each individual data point for an alarm. It is expressed in seconds.
 - **Evaluation Period** is the number of the most recent periods, or data points, to evaluate when determining alarm state.
 - **Datapoints to Alarm** is the number of data points within the evaluation period that must be breaching to cause the alarm to go to the *ALARM* state. The breaching data points do not have to be consecutive, they just must all be within the last number of data points equal to **Evaluation Period**.



- For each alarm, you can specify CloudWatch to treat missing data points as any of the following:
 - *missing*—the alarm does not consider missing data points when evaluating whether to change state (default)
 - *notBreaching*—missing data points are treated as being within the threshold
 - *breaching*—missing data points are treated as breaching the threshold
 - *ignore*—the current alarm state is maintained
- You can now create tags in CloudWatch alarms that let you define policy controls for your AWS resources. This enables you to create resource level policies for your alarms.

CloudWatch Dashboard

- Customizable home pages in the CloudWatch console that you can use to monitor your resources in a single view, even those spread across different regions.
- There is no limit on the number of CloudWatch dashboards you can create.
- All dashboards are **global**, not region-specific.
- You can add, remove, resize, move, edit or rename a graph. You can metrics manually in a graph.

EventBridge

- Deliver near real-time stream of system events that describe changes in AWS resources.
- Events respond to these operational changes and take corrective action as necessary, by sending messages to respond to the environment, activating functions, making changes, and capturing state information.



- Concepts
 - **Events** - indicates a change in your AWS environment.
 - **Targets** - processes events.
 - **Rules** - matches incoming events and routes them to targets for processing.

CloudWatch Logs

- Features
 - Monitor logs from EC2 instances in real-time
 - Monitor CloudTrail logged events
 - By default, logs are kept indefinitely and never expire
 - Archive log data
 - Log Route 53 DNS queries
- **CloudWatch Logs Insights** enables you to interactively search and analyze your log data in CloudWatch Logs using queries.
- **CloudWatch Vended logs** are logs that are natively published by AWS services on behalf of the customer. **VPC Flow logs** is the first Vended log type that will benefit from this tiered model.
- After the CloudWatch Logs agent begins publishing log data to Amazon CloudWatch, you can search and filter the log data by creating one or more metric filters. **Metric filters** define the terms and patterns to look for in log data as it is sent to CloudWatch Logs.
- Filters **do not** retroactively filter data. Filters only publish the metric data points for events that happen after the filter was created. Filtered results return the first 50 lines, which will not be displayed if the timestamp on the filtered results is earlier than the metric creation time.
- Metric Filter Concepts
 - filter pattern - you use the pattern to specify what to look for in the log file.
 - metric name - the name of the CloudWatch metric to which the monitored log information should be published.
 - metric namespace - the destination namespace of the new CloudWatch metric.
 - metric value - the numerical value to publish to the metric each time a matching log is found.
 - default value - the value reported to the metric filter during a period when no matching logs are found. By setting this to 0, you ensure that data is reported during every period.
- You can create two subscription filters with different filter patterns on a single log group.

CloudWatch Agent

- Collect more logs and system-level metrics from EC2 instances and your on-premises servers.
- Needs to be installed.

Authentication and Access Control

- Use IAM users or roles for authenticating who can access
- Use Dashboard Permissions, IAM identity-based policies, and service-linked roles for managing access control.
- A *permissions policy* describes who has access to what.
 - Identity-Based Policies
 - Resource-Based Policies



- There are no CloudWatch Amazon Resource Names (ARNs) for you to use in an IAM policy. Use an * (asterisk) instead as the resource when writing a policy to control access to CloudWatch actions.

Pricing

- You are charged for the number of metrics you have per month
- You are charged per 1000 metrics requested using CloudWatch API calls
- You are charged per dashboard per month
- You are charged per alarm metric (Standard Resolution and High Resolution)
- You are charged per GB of collected, archived and analyzed log data
- There is no Data Transfer IN charge, only Data Transfer Out.
- You are charged per million custom events and per million cross-account events
- Logs Insights is priced per query and charges based on the amount of ingested log data scanned by the query.

References:

<https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring>

<https://aws.amazon.com/cloudwatch/faqs/>

AWS Lambda

- A serverless compute service.
- Lambda executes your code only when needed and scales automatically.
- Lambda functions are stateless - no affinity to the underlying infrastructure.
- You choose the amount of memory you want to allocate to your functions and AWS Lambda allocates proportional CPU power, network bandwidth, and disk I/O.
- AWS Lambda is SOC, HIPAA, PCI, ISO compliant.
- Natively supports the following languages:
 - Node.js
 - Java
 - C#
 - Go
 - Python
 - Ruby
 - PowerShell
- You can also provide your own custom runtime.

Components of a Lambda Application

- **Function** – a script or program that runs in Lambda. Lambda passes invocation events to your function. The function processes an event and returns a response.



- **Runtimes** – Lambda runtimes allow functions in different languages to run in the same base execution environment. The runtime sits in-between the Lambda service and your function code, relaying invocation events, context information, and responses between the two.
- **Layers** – Lambda layers are a distribution mechanism for libraries, custom runtimes, and other function dependencies. Layers let you manage your in-development function code independently from the unchanging code and resources that it uses.
- **Event source** – an AWS service or a custom service that triggers your function and executes its logic.
- **Downstream resources** – an AWS service that your Lambda function calls once it is triggered.
- **Log streams** – While Lambda automatically monitors your function invocations and reports metrics to CloudWatch, you can annotate your function code with custom logging statements that allow you to analyze the execution flow and performance of your Lambda function.
- AWS Serverless Application Model

Lambda Functions

- You upload your application code in the form of one or more *Lambda functions*. Lambda stores code in Amazon S3 and encrypts it at rest.
- To create a Lambda function, you first package your code and dependencies in a deployment package. Then, you upload the deployment package to create your Lambda function.
- After your Lambda function is in production, Lambda automatically monitors functions on your behalf, reporting metrics through Amazon CloudWatch.
- Configure **basic function settings** including the description, memory usage, execution timeout, and role that the function will use to execute your code.
- **Environment variables** are always encrypted at rest, and can be encrypted in transit as well.
- **Versions and aliases** are secondary resources that you can create to manage function deployment and invocation.
- A **layer** is a ZIP archive that contains libraries, a custom runtime, or other dependencies. Use layers to manage your function's dependencies independently and keep your deployment package small.
- You can configure a function to mount an Amazon EFS file system to a local directory. With Amazon EFS, your function code can access and modify shared resources securely and at high concurrency.

Invoking Functions

- Lambda supports **synchronous** and **asynchronous invocation** of a Lambda function. You can control the invocation type only when you invoke a Lambda function (referred to as *on-demand invocation*).
- An **event source** is the entity that publishes events, and a Lambda function is the custom code that processes the events.
- *Event source mapping* maps an event source to a Lambda function. It enables automatic invocation of your Lambda function when events occur.
- Lambda provides event source mappings for the following services.
 - Amazon Kinesis
 - Amazon DynamoDB



- Amazon Simple Queue Service
- Your functions' *concurrency* is the number of instances that serve requests at a given time. When your function is invoked, Lambda allocates an instance of it to process the event. When the function code finishes running, it can handle another request. If the function is invoked again while a request is still being processed, another instance is allocated, which increases the function's concurrency.
- To ensure that a function can always reach a certain level of concurrency, you can configure the function with **reserved concurrency**. When a function has reserved concurrency, no other function can use that concurrency. Reserved concurrency also limits the maximum concurrency for the function.
- To enable your function to scale without fluctuations in latency, use **provisioned concurrency**. By allocating provisioned concurrency before an increase in invocations, you can ensure that all requests are served by initialized instances with very low latency.

Configuring a Lambda Function to Access Resources in a VPC

In AWS Lambda, you can set up your function to establish a connection to your virtual private cloud (VPC). With this connection, your function can access the private resources of your VPC during execution like EC2, RDS and many others.

By default, AWS executes your Lambda function code securely within a VPC. Alternatively, you can enable your Lambda function to access resources inside your private VPC by providing additional VPC-specific configuration information such as VPC subnet IDs and security group IDs. It uses this information to set up elastic network interfaces which enable your Lambda function to connect securely to other resources within your VPC.

Lambda@Edge

- Lets you run Lambda functions to customize content that CloudFront delivers, executing the functions in AWS locations closer to the viewer. The functions run in response to CloudFront events, without provisioning or managing servers.
- You can use Lambda functions to change CloudFront requests and responses at the following points:
 - After CloudFront receives a request from a viewer (viewer request)
 - Before CloudFront forwards the request to the origin (origin request)
 - After CloudFront receives the response from the origin (origin response)
 - Before CloudFront forwards the response to the viewer (viewer response)
- You can automate your serverless application's release process using AWS CodePipeline and AWS CodeDeploy.
- Lambda will automatically track the behavior of your Lambda function invocations and provide feedback that you can monitor. In addition, it provides metrics that allows you to analyze the full function invocation spectrum, including event source integration and whether downstream resources perform as expected.



Pricing

- You are charged based on the total number of requests for your functions and the duration, the time it takes for your code to execute.

References:

<https://docs.aws.amazon.com/lambda/latest/dg>

<https://aws.amazon.com/lambda/faqs/>

AWS Elastic Beanstalk

- Allows you to quickly deploy and manage applications in the AWS Cloud without worrying about the infrastructure that runs those applications.
- Elastic Beanstalk automatically handles the details of capacity provisioning, load balancing, scaling, and application health monitoring for your applications.
- It is a Platform-as-a-Service
- Elastic Beanstalk supports the following languages:
 - Go
 - Java
 - .NET
 - Node.js
 - PHP
 - Python
 - Ruby
- Elastic Beanstalk supports the following web containers:
 - Tomcat
 - Passenger
 - Puma
- Elastic Beanstalk supports Docker containers.
- Your application's domain name is in the format:
`subdomain.region.elasticbeanstalk.com`

Environment Pages

- The **Configuration** page shows the resources provisioned for this environment. This page also lets you configure some of the provisioned resources.
- The **Health** page shows the status and detailed health information about the EC2 instances running your application.
- The **Monitoring** page shows the statistics for the environment, such as average latency and CPU utilization. You also use this page to create alarms for the metrics that you are monitoring.



- The **Events** page shows any informational or error messages from services that this environment is using.
- The **Tags** page shows tags — key-value pairs that are applied to resources in the environment. You use this page to manage your environment's tags.

Elastic Beanstalk Concepts

- **Application** - a logical collection of Elastic Beanstalk components, including environments, versions, and environment configurations. It is conceptually similar to a folder.
- **Application Version** - refers to a specific, labeled iteration of deployable code for a web application. An application version points to an Amazon S3 object that contains the deployable code. Applications can have many versions and each application version is unique.
- **Environment** - a version that is deployed on to AWS resources. Each environment runs only a single application version at a time, however you can run the same version or different versions in many environments at the same time.
- **Environment Tier** - determines whether Elastic Beanstalk provisions resources to support an application that handles HTTP requests or an application that pulls tasks from a queue. An application that serves HTTP requests runs in a **web server environment**. An environment that pulls tasks from an Amazon SQS queue runs in a **worker environment**.
- **Environment Configuration** - identifies a collection of parameters and settings that define how an environment and its associated resources behave.
- **Configuration Template** - a starting point for creating unique environment configurations.
- There is a limit to the number of application versions you can have. You can avoid hitting the limit by applying an *application version lifecycle policy* to your applications to tell Elastic Beanstalk to delete application versions that are old, or to delete application versions when the total number of versions for an application exceeds a specified number.

Environment Types

- **Load-balancing, Autoscaling Environment** - automatically starts additional instances to accommodate increasing load on your application.
- **Single-Instance Environment** - contains one Amazon EC2 instance with an Elastic IP address.

Environment Configurations

- Your environment contains:
 - Your **EC2 virtual machines** configured to run web apps on the platform that you choose.
 - An **Auto Scaling group** that ensures that there is always one instance running in a single-instance environment, and allows configuration of the group with a range of instances to run in a load-balanced environment.
 - When you enable load balancing, Elastic Beanstalk creates an **Elastic Load Balancing load balancer** to distributes traffic among your environment's instances.



- Elastic Beanstalk provides integration with **Amazon RDS** to help you add a database instance to your Elastic Beanstalk environment : **MySQL, PostgreSQL, Oracle, or SQL Server**. When you add a database instance to your environment, Elastic Beanstalk provides connection information to your application by setting environment properties for the database hostname, port, user name, password, and database name.
- You can use **environment properties** to pass secrets, endpoints, debug settings, and other information to your application. Environment properties help you run your application in multiple environments for different purposes, such as development, testing, staging, and production.
- You can configure your environment to use **Amazon SNS** to notify you of important events that affect your application.
- Your environment is available to users at a **subdomain of elasticbeanstalk.com**. When you create an environment, you can choose a unique subdomain that represents your application.

Monitoring

- Elastic Beanstalk Monitoring console displays your environment's status and application health at a glance.
- Elastic Beanstalk reports the health of a web server environment depending on how the application running in it responds to the health check.
- **Enhanced health reporting** is a feature that you can enable on your environment to allow AWS Elastic Beanstalk to gather additional information about resources in your environment. Elastic Beanstalk analyzes the information gathered to provide a better picture of overall environment health and aid in the identification of issues that can cause your application to become unavailable.
- You can create alarms for metrics to help you monitor changes to your environment so that you can easily identify and mitigate problems before they occur.
- EC2 instances in your Elastic Beanstalk environment generate logs that you can view to troubleshoot issues with your application or configuration files.

Security

- When you create an environment, Elastic Beanstalk prompts you to provide two AWS IAM roles: a **service role** and an **instance profile**.
 - Service Roles - assumed by Elastic Beanstalk to use other AWS services on your behalf.
 - Instance Profiles - applied to the instances in your environment and allows them to retrieve application versions from S3, upload logs to S3, and perform other tasks that vary depending on the environment type and platform.
- User Policies - allow users to create and manage Elastic Beanstalk applications and environments.

Pricing



- There is no additional charge for Elastic Beanstalk. You pay only for the underlying AWS resources that your application consumes.

References:

<https://docs.aws.amazon.com/elasticbeanstalk/latest/dg>

<https://aws.amazon.com/elasticbeanstalk/faqs/>

AWS Storage Gateway

- The service enables **hybrid storage** between on-premises environments and the AWS Cloud.
- It integrates on-premises enterprise applications and workflows with Amazon's block and object cloud storage services through industry standard storage protocols.
- The service stores files as native S3 objects, archives virtual tapes in Amazon Glacier, and stores EBS Snapshots generated by the Volume Gateway with Amazon EBS.

Storage Solutions

- **File Gateway** - supports a file interface into S3 and combines a service and a virtual software appliance.
 - The software appliance, or gateway, is deployed into your on-premises environment as a virtual machine running on VMware ESXi or Microsoft Hyper-V hypervisor.
 - File gateway supports
 - S3 Standard
 - S3 Standard - Infrequent Access
 - S3 One Zone - IA
 - With a file gateway, you can do the following:
 - You can store and retrieve files directly using the NFS version 3 or 4.1 protocol.
 - You can store and retrieve files directly using the SMB file system version, 2 and 3 protocol.
 - You can access your data directly in S3 from any AWS Cloud application or service.
 - You can manage your S3 data using lifecycle policies, cross-region replication, and versioning.
 - File Gateway now supports Amazon S3 Object Lock, enabling write-once-read-many (WORM) file-based systems to store and access objects in Amazon S3.
 - Any modifications such as file edits, deletes or renames from the gateway's NFS or SMB clients are stored as new versions of the object, without overwriting or deleting previous versions.
- **Volume Gateway** - provides cloud-backed storage volumes that you can mount as iSCSI devices from your on-premises application servers.



- **Cached volumes** – you store your data in S3 and retain a copy of frequently accessed data subsets locally. Cached volumes can range from 1 GiB to 32 TiB in size and must be rounded to the nearest GiB. Each gateway configured for cached volumes can support up to 32 volumes.
- **Stored volumes** – if you need low-latency access to your entire dataset, first configure your on-premises gateway to store all your data locally. Then asynchronously back up point-in-time snapshots of this data to S3. Stored volumes can range from 1 GiB to 16 TiB in size and must be rounded to the nearest GiB. Each gateway configured for stored volumes can support up to 32 volumes.
- AWS Storage Gateway customers using the Volume Gateway configuration for block storage can detach and attach volumes, from and to a Volume Gateway. You can use this feature to migrate volumes between gateways to refresh underlying server hardware, switch between virtual machine types, and move volumes to better host platforms or newer Amazon EC2 instances.
- **Tape Gateway** - archive backup data in Amazon Glacier.
 - Has a virtual tape library (VTL) interface to store data on virtual tape cartridges that you create.
 - Deploy your gateway on an EC2 instance to provision iSCSI storage volumes in AWS.
 - The AWS Storage Gateway service integrates Tape Gateway with Amazon S3 Glacier Deep Archive storage class, allowing you to store virtual tapes in the lowest-cost Amazon S3 storage class.
 - Tape Gateway also has the capability to move your virtual tapes archived in Amazon S3 Glacier Flexible Retrieval to Amazon S3 Glacier Deep Archive storage class, enabling you to further reduce the monthly cost to store long-term data in the cloud by up to 75%.

Storage Gateway Hosting Options

- As a VM containing the Storage Gateway software, run on VMware ESXi, Microsoft Hyper-V on premises
- As a VM in VMware Cloud on AWS
- As a hardware appliance on premises
- As an AMI in an EC2 instance

Storage Gateway stores volume, snapshot, tape, and file data in the AWS Region in which your gateway is activated. File data is stored in the AWS Region where your S3 bucket is located.

The local gateway appliance maintains a cache of recently written or read data so your applications can have low-latency access to data that is stored durably in AWS. The gateways use a **read-through and write-back** cache.

File Gateway File Share

- You can create an NFS or SMB file share using the AWS Management Console or service API.



- After your file gateway is activated and running, you can add additional file shares and grant access to S3 buckets.
- You can use a file share to access objects in an S3 bucket that belongs to a different AWS account.
- The AWS Storage Gateway service added support for Access Control Lists (ACLs) to Server Message Block (SMB) shares on the File Gateway, helping enforce data security standards when using the gateway for storing and accessing data in Amazon Simple Storage Service (S3).
- After your file gateway is activated and running, you can add additional file shares and grant access to S3 buckets.

Security

- You can use AWS KMS to encrypt data written to a virtual tape.
- Storage Gateway uses Challenge-Handshake Authentication Protocol (CHAP) to authenticate iSCSI and initiator connections. CHAP provides protection against playback attacks by requiring authentication to access storage volume targets.
- Authentication and access control with IAM.

Compliance

- Storage Gateway is HIPAA eligible.
- Storage Gateway in compliance with the Payment Card Industry Data Security Standard (PCI DSS)

Pricing

- You are charged based on the type and amount of storage you use, the requests you make, and the amount of data transferred out of AWS.
- You are charged only for the amount of data you write to the Tape Gateway tape, not the tape capacity.

References:

<https://docs.aws.amazon.com/storagegateway/latest/userguide/>
<https://aws.amazon.com/storagegateway/faqs/>



Amazon ElastiCache

- ElastiCache is a distributed **in-memory cache** environment in the AWS Cloud.
- ElastiCache works with both the **Redis** and **Memcached** engines.

Components

- ElastiCache Nodes
 - A **node** is a fixed-size chunk of secure, network-attached RAM. A node can exist in isolation from or in some relationship to other nodes.
 - Every node within a cluster is the same instance type and runs the same cache engine. Each cache node has its own Domain Name Service (DNS) name and port.
- If a maintenance event is scheduled for a given week, it will be initiated and completed at some point during the 60 minute maintenance window you specify.
- ElastiCache can be used for storing session state.
- ElastiCache Redis
 - Existing applications that use Redis can use ElastiCache with almost no modification.
 - Features
 - Automatic detection and recovery from cache node failures.
 - Multi-AZ with automatic failover of a failed primary cluster to a read replica in Redis clusters that support replication.
 - Redis (cluster mode enabled) supports partitioning your data across up to 250 shards.
 - Redis supports in-transit and at-rest encryption with authentication so you can build HIPAA-compliant applications.
 - Flexible Availability Zone placement of nodes and clusters for increased fault tolerance.
 - Data is persistent.
 - Can be used as a datastore.
 - Not multi-threaded.
 - Amazon ElastiCache for Redis supports self-service updates, which allows you to apply service updates at the time of your choosing and track the progress in real-time.
 - Cache data if:
 - It is slow or expensive to acquire when compared to cache retrieval.
 - It is accessed with sufficient frequency.
 - It is relatively static, or if rapidly changing, staleness is not a significant issue.
 - **Redis sorted sets** guarantee both uniqueness and element ordering. Each time a new element is added to the sorted set it's reranked in real time. It's then added to the set in its appropriate numeric position.
 - In the **Redis publish/subscribe** paradigm, you send a message to a specific channel not knowing who, if anyone, receives it. Recipients of the message are those who are subscribed to the channel.



- **Redis hashes** are hashes that map string names to string values.
- Components
 - **Redis Shard** - a grouping of one to six related nodes. A Redis (cluster mode disabled) cluster always has one shard. A Redis (cluster mode enabled) cluster can have 1–90 shards.
 - A *multiple node shard* implements replication by have one read/write primary node and 1–5 replica nodes.
 - If there is more than one node in a shard, the shard supports replication with one node being the read/write primary node and the others read-only replica nodes.
 - **Redis Cluster** - a logical grouping of one or more ElastiCache for Redis Shards. Data is partitioned across the shards in a Redis (cluster mode enabled) cluster.
- For improved fault tolerance, have at least two nodes in a Redis cluster and enabling **Multi-AZ with automatic failover**.
- Replica nodes use asynchronous replication mechanisms to keep synchronized with the primary node.
- If any primary has no replicas and the primary fails, you lose all that primary's data.
- You can use backup and restore to **migrate** to Redis (cluster mode enabled) and resize your Redis (cluster mode enabled).
- Redis (cluster mode disabled) vs Redis (cluster mode enabled)

	Redis (cluster mode disabled)	Redis (cluster mode enabled)
Shards (node groups)	1	1-90
Replicas for each shard (node group)	0-5	0-5
Data partitioning	No	Yes
Add/ Delete replicas	Yes	Yes
Add/ Delete node groups	No	No
Supports scale up	Yes	No
Supports engine upgrades	Yes	Yes
Promote replica to primary	Yes	No
Multi-AZ with automatic failover	Yes, with at least a replica. Optional. On by default.	Required
Backup/ Restore	Yes	Yes



- You can vertically scale up or scale down your sharded Redis Cluster on demand. Amazon ElastiCache resizes your cluster by changing the node type, while the cluster continues to stay online and serve incoming requests.
- You can set up automatic snapshots or initiate manual backups, and then seed new ElastiCache for Redis clusters. You can also export your snapshots to an S3 bucket of your choice for disaster recovery, analysis or cross-region backup and restore.
- Endpoints
 - **Single Node Redis (cluster mode disabled)** Endpoints - used to connect to the cluster for both reads and writes.
 - **Multi-Node Redis (cluster mode disabled)** Endpoints - use the primary endpoint for all writes to the cluster. The read endpoint points to your read replicas.
 - **Redis (cluster mode enabled)** Endpoints - has a single configuration endpoint. By connecting to the configuration endpoint, your application is able to discover the primary and read endpoints for each shard in the cluster.
- Parameter Groups
 - **Cache parameter group** is a named collection of engine-specific parameters that you can apply to a cluster.
 - Parameters are used to control memory usage, eviction policies, item sizes, and more.
- Redis Security
 - ElastiCache for Redis node access is restricted to applications running on whitelisted EC2 instances. You can control access of your cluster by using subnet groups or security groups. By default, network access to your clusters is turned off.
 - By default, all new ElastiCache for Redis clusters are launched in a VPC environment. Use subnet groups to grant cluster access from Amazon EC2 instances running on specific subnets.
 - ElastiCache for Redis supports TLS and in-place encryption for nodes running specified versions of the ElastiCache for Redis engine.
 - You can use your own AWS KMS Keys to encrypt data at rest in ElastiCache for Redis.
- Redis Backups
 - A point-in-time copy of a Redis cluster.
 - Backups consist of all the data in a cluster plus some metadata.
- Global Datastore
 - A new feature that provides fully managed, secure cross-region replication. You can now write to your ElastiCache for Redis cluster in one region and have the data available for reading in two other cross-region replica clusters.
 - In the unlikely event of regional degradation, one of the healthy cross-region replica clusters can be promoted to become the primary cluster with full read/write capabilities.

ElastiCache Memcached

- Features



- Automatic detection and recovery from cache node failures.
- Automatic discovery of nodes within a cluster enabled for automatic discovery, so that no changes need to be made to your application when you add or remove nodes.
- Flexible Availability Zone placement of nodes and clusters.
- **ElastiCache Auto Discovery** feature for Memcached lets your applications identify all of the nodes in a cache cluster and connect to them.
- ElastiCache node access is restricted to applications running on whitelisted EC2 instances. You can control the instances that can access your cluster by using subnet groups or security groups.
- It is not persistent.
- Supports large nodes with multiple cores or threads.
- Does not support multi-AZ failover or replication
- Does not support snapshots
- Components
 - **Memcached cluster** - a logical grouping of one or more ElastiCache Nodes. Data is partitioned across the nodes in a Memcached cluster.
 - Memcached supports up to 100 nodes per customer for each Region with each cluster having 1–20 nodes.
 - When you partition your data, use *consistent hashing*.
 - **Endpoint** - the unique address your application uses to connect to an ElastiCache node or cluster.
 - Each node in a Memcached cluster has its own endpoint.
 - The cluster also has an endpoint called the *configuration endpoint*.
 - **ElastiCache parameter group** - a named collection of engine-specific parameters that you can apply to a cluster. Parameters are used to control memory usage, eviction policies, item sizes, and more.
 - ElastiCache allows you to control access to your clusters using **security groups**. By default, network access to your clusters is turned off.
 - A **subnet group** is a collection of subnets that you can designate for your clusters running in a VPC environment. If you create a cluster in a VPC, then you must specify a *cache subnet group*. ElastiCache uses that cache subnet group to choose a subnet and IP addresses within that subnet to associate with your cache nodes.
- Mitigating Failures
 - Node Failures
 - Spread your cached data over more nodes. Because Memcached does not support replication, a node failure will always result in some data loss from your cluster.
 - Availability Zone Failure
 - Locate your nodes in as many Availability Zones as possible. In the unlikely event of an AZ failure, you will lose the data cached in that AZ, not the data cached in the other AZs.



- ElastiCache uses DNS entries to allow client applications to locate servers (nodes). The DNS name for a node remains constant, but the IP address of a node can change over time.

Caching Strategies

- **Lazy Loading** - a caching strategy that loads data into the cache only when necessary.
 - Only requested data is cached.
 - Node failures are not fatal.
 - There is a cache miss penalty.
 - Stale data.
- **Write Through** - adds data or updates data in the cache whenever data is written to the database.
 - Data in the cache is never stale.
 - Write penalty vs. Read penalty. Every write involves two trips: A write to the cache and a write to the database.
 - Missing data.
 - Cache churn.
- By adding a time to live (TTL) value to each write, we are able to enjoy the advantages of each strategy and largely avoid cluttering up the cache with superfluous data.

Scaling ElastiCache for Memcached Clusters

- Scaling Memcached Horizontally
 - The Memcached engine supports partitioning your data across multiple nodes. Because of this, Memcached clusters scale horizontally easily. A Memcached cluster can have 1 to 20 nodes. To horizontally scale your Memcached cluster, just add or remove nodes.
- Scaling Memcached Vertically
 - When you scale your Memcached cluster up or down, you must create a new cluster. Memcached clusters always start out empty unless your application populates it.

Monitoring

- The service continuously monitors the health of your instances. In case a node experiences failure or a prolonged degradation in performance, ElastiCache will automatically restart the node and associated processes.
- ElastiCache provides both host-level metrics and metrics that are specific to the cache engine software. These metrics are measured and published for each Cache node in 60-second intervals.
- Monitor events with **ElastiCache Events**. When significant events happen on a cache cluster, including failure to add a node, success in adding a node, the modification of a security group, and others, ElastiCache sends a notification to a specific SNS topic.
- Monitor costs with tags.



Redis VS Memcached

- Memcached is designed for **simplicity** while Redis offers a **rich set of features** that make it effective for a wide range of use cases.

	Redis (cluster mode enabled)	Redis (cluster mode disabled)	Memcached
Data Types	string, sets, sorted sets, lists, hashes, bitmaps, hyperloglog, geospatial indexes	string, sets, sorted sets, lists, hashes, bitmaps, hyperloglog, geospatial indexes	string, objects (like databases)
Data Partitioning (distribute your data among multiple nodes)	Supported	Unsupported	Supported
Modifiable cluster	Only versions 3.2.10 and later	Yes	Yes
Online resharding	Only versions 3.2.10 and later	No	No
Encryption	3.2.6, 4.0.10 and later	3.2.6, 4.0.10 and later	Unsupported
Sub-millisecond latency	Yes	Yes	Yes
FedRAMP, PCI DSS and HIPAA compliant	3.2.6, 4.0.10 and later	3.2.6, 4.0.10 and later	No
Multi-threaded (make use of multiple processing cores)	No	No	No
Node type upgrading	No	Yes	No
Engine upgrading	Yes		
Cluster replication (create multiple copies of a primary cluster)	Supported	Supported	Unsupported
Multi-AZ for automatic failover	Required	Optional	Unsupported
Transactions (execute a group of commands as an isolated and atomic operation)	Supported	Supported	Unsupported
Pub/Sub capability	Yes	Yes	No



Backup and restore (keep your data on disk with a point in time snapshot)	Supported	Supported	Unsupported
Lua Scripting (execute transactional Lua scripts)	Supported	Supported	Unsupported
Use case	- You need to partition your data across two to 250 or 500 nodes if the Redis engine version is 5.0.6 or higher.(clustered mode only). - You need geospatial indexing (clustered mode or non-clustered mode). - You don't need to support multiple databases. - Plus features of non-clustered mode.	- You need complex data types, such as strings, hashes, lists, sets, sorted sets, and bitmaps. - You need to sort or rank in-memory datasets. - You need persistence of your key store. - You need to replicate your data from the primary to one or more read replicas for read intensive applications. - You need automatic failover if your primary node fails. - You need pub/sub capabilities. - You need backup and restore capabilities. - You need to support multiple databases.	- You need the simplest model possible. - You need to run large nodes with multiple cores or threads. - You need the ability to scale out and in, adding and removing nodes as demand on your system increases and decreases. - You need to cache objects, such as a database. - Needs Auto Discovery to simplify the way an application connects to a cluster.



Pricing

- With on-demand nodes you pay only for the resources you consume by the hour without any long-term commitments.
- With Reserved Nodes, you can make a low, one-time, up-front payment for each node you wish to reserve for a 1 or 3 year term. In return, you receive a significant discount off the ongoing hourly usage rate for the Node(s) you reserve.
- ElastiCache provides storage space for one snapshot free of charge for each active ElastiCache for Redis cluster. Additional backup storage is charged.
- EC2 Regional Data Transfer charges apply when transferring data between an EC2 instance and an ElastiCache Node in different Availability Zones of the same Region.

References:

<https://aws.amazon.com/elasticache/redis-details/>

<https://aws.amazon.com/elasticache/redis-vs-memcached/>

<https://aws.amazon.com/elasticache/features/>

Amazon DynamoDB

- NoSQL database service that provides fast and predictable performance with seamless scalability.
- Offers encryption at rest.
- You can create database tables that can store and retrieve any amount of data, and serve any level of request traffic.
- You can scale up or scale down your tables' throughput capacity without downtime or performance degradation, and use the AWS Management Console to monitor resource utilization and performance metrics.
- Provides on-demand backup capability as well as enable point-in-time recovery for your DynamoDB tables. With point-in-time recovery, you can restore that table to any point in time during the **last 35 days**.
- All of your data is stored in partitions, backed by solid state disks (SSDs) and automatically replicated across multiple AZs in an AWS region, providing built-in high availability and data durability.
- You can create tables that are automatically replicated across two or more AWS Regions, with full support for multi-master writes.
- AWS now specifies the IP address ranges for Amazon DynamoDB endpoints. You can use these IP address ranges in your routing and firewall policies to control outbound application traffic. You can also use these ranges to control outbound traffic for applications in your Amazon Virtual Private Cloud, behind AWS Virtual Private Network or AWS Direct Connect.

Core Components

- **Tables** - a collection of items



- DynamoDB stores data in a table, which is a collection of data.
- Are schemaless.
- There is an initial limit of 256 tables per region.
- **Items** - a collection of attributes
 - DynamoDB uses **primary keys** to uniquely identify each item in a table and **secondary indexes** to provide more querying flexibility.
 - Each table contains zero or more items.
- **Attributes** - a fundamental data element
 - DynamoDB supports nested attributes up to 32 levels deep.
- **Primary Key** - uniquely identifies each item in the table, so that no two items can have the same key. Must be scalar.
 - **Partition key** - a simple primary key, composed of one attribute.
 - **Partition key and sort key** (*composite primary key*) - composed of two attributes.
 - DynamoDB uses the partition key value as input to an internal hash function. The output from the hash function determines the partition in which the item will be stored. All items with the same partition key are stored together, in sorted order by sort key value. If no sort key is used, no two items can have the same partition key value.
- **Secondary Indexes** - lets you query the data in the table using an alternate key, in addition to queries against the primary key.
 - You can create one or more secondary indexes on a table.
 - Two kinds of indexes:
 - **Global secondary index** – An index with a partition key and sort key that can be different from those on the table.
 - **Local secondary index** – An index that has the same partition key as the table, but a different sort key.
 - You can define up to 20 global secondary indexes and 5 local secondary indexes per table.
- **DynamoDB Streams** - an optional feature that captures data modification events in DynamoDB tables.
 - The naming convention for DynamoDB Streams endpoints is *streams.dynamodb..amazonaws.com*
 - Each event is represented by a *stream record*, and captures the following events:
 - A new item is added to the table: captures an image of the entire item, including all of its attributes.
 - An item is updated: captures the "before" and "after" image of any attributes that were modified in the item.
 - An item is deleted from the table: captures an image of the entire item before it was deleted.
 - Each stream record also contains the name of the table, the event timestamp, and other metadata.
 - Stream records are organized into groups, or **shards**. Each shard acts as a container for multiple stream records, and contains information required for accessing and iterating through these records.



- Stream records have a lifetime of 24 hours; after that, they are automatically removed from the stream.
- You can use DynamoDB Streams together with AWS Lambda to create a *trigger*, which is a code that executes automatically whenever an event of interest appears in a stream.
- DynamoDB Streams enables powerful solutions such as data replication within and across Regions, materialized views of data in DynamoDB tables, data analysis using Kinesis materialized views, and much more.

Data Types for Attributes

- **Scalar Types** – A scalar type can represent exactly one value. The scalar types are number, string, binary, Boolean, and null. Primary keys should be scalar types.
- **Document Types** – A document type can represent a complex structure with nested attributes—such as you would find in a JSON document. The document types are list and map.
- **Set Types** – A set type can represent multiple scalar values. The set types are string set, number set, and binary set.

Other Notes:

- When you read data from a DynamoDB table, the response might not reflect the results of a recently completed write operation. The response might include some stale data, but you should **eventually have consistent reads**.
- When you request a **strongly consistent read**, DynamoDB returns a response with the most up-to-date data, reflecting the updates from all prior write operations that were successful. A strongly consistent read might not be available if there is a network delay or outage.
- DynamoDB does not support strongly consistent reads across AWS regions
- When you create a table or index in DynamoDB, you must specify your throughput capacity requirements for read and write activity in terms of:
 - One **read capacity unit** represents one strongly consistent read per second, or two eventually consistent reads per second, for an item up to 4 KB in size. If you need to read an item that is larger than 4 KB, DynamoDB will need to consume additional read capacity units.
 - One **write capacity unit** represents one write per second for an item up to 1 KB in size. If you need to write an item that is larger than 1 KB, DynamoDB will need to consume additional write capacity units.
- *Throttling* prevents your application from consuming too many capacity units. DynamoDB can throttle read or write requests that exceed the throughput settings for a table, and can also throttle read requests exceeds for an index.
- When a request is throttled, it fails with an **HTTP 400** code (Bad Request) and a *ProvisionedThroughputExceededException*.

Throughput Management



- Provisioned throughput - manually defined maximum amount of capacity that an application can consume from a table or index. If your application exceeds your provisioned throughput settings, it is subject to request throttling. Free tier eligible.
 - DynamoDB auto scaling
 - Define a range (upper and lower limits) for **read and write capacity units**, and define a target utilization percentage within that range.
 - A table or a global secondary index can increase its **provisioned read and write capacity** to handle sudden increases in traffic, without request throttling.
 - DynamoDB auto scaling can decrease the throughput when the workload decreases so that you don't pay for unused provisioned capacity.
 - Reserved capacity - with reserved capacity, you pay a one-time upfront fee and commit to a minimum usage level over a period of time, for cost-saving solutions.
- Amazon DynamoDB on-demand is a flexible capacity mode for DynamoDB capable of serving thousands of requests per second without capacity planning. When you choose on-demand capacity mode, DynamoDB instantly accommodates your workloads as they ramp up or down to any previously reached traffic level. If a workload's traffic level hits a new peak, DynamoDB adapts rapidly to accommodate the workload. DynamoDB on-demand offers simple pay-per-request pricing for read and write requests so that you only pay for what you use, making it easy to balance costs and performance.

Capacity Unit Consumption

- CUC for Reads - strongly consistent read request consumes one read capacity unit, while an eventually consistent read request consumes 0.5 of a read capacity unit.
 - GetItem - reads a single item from a table.
 - BatchGetItem - reads up to 100 items, from one or more tables.
 - Query - reads multiple items that have the same partition key value.
 - Scan - reads all of the items in a table
- CUC for Writes
 - PutItem - writes a single item to a table.
 - UpdateItem - modifies a single item in the table.
 - DeleteItem - removes a single item from a table.
 - BatchWriteItem - writes up to 25 items to one or more tables.
 - Calculating the Required Read and Write Capacity Unit for Your DynamoDB table:
<https://tutorialsdojo.com/calculating-the-required-read-and-write-capacity-unit-for-your-dynamo-db-table/>

DynamoDB Auto Scaling

- When you use the AWS Management Console to create a new table, DynamoDB auto scaling is enabled for that table by default.



- Uses the AWS Application Auto Scaling service to dynamically adjust provisioned throughput capacity on your behalf, in response to actual traffic patterns.
- You create a *scaling policy* for a table or a global secondary index. The scaling policy specifies whether you want to scale read capacity or write capacity (or both), and the minimum and maximum provisioned capacity unit settings for the table or index. The scaling policy also contains a *target utilization*, which is the percentage of consumed provisioned throughput at a point in time.
- DynamoDB auto scaling doesn't prevent you from manually modifying provisioned throughput settings.
- If you enable DynamoDB auto scaling for a table that has one or more global secondary indexes, AWS highly recommends that you also apply auto scaling uniformly to those indexes.

Tagging

- Tags can help you:
 - Quickly identify a resource based on the tags you've assigned to it.
 - See AWS bills broken down by tags.
- Each DynamoDB table can have only one tag with the same key. If you try to add an existing tag (same key), the existing tag value will be updated to the new value.
- Maximum number of tags per resource: 50

DynamoDB Items

- You can use the *UpdateItem* operation to implement an **atomic counter** - a numeric attribute that is incremented, unconditionally, without interfering with other write requests.
- DynamoDB optionally supports conditional writes for these operations: *PutItem*, *UpdateItem*, *DeleteItem*. A conditional write will succeed only if the item attributes meet one or more expected conditions.
- Conditional writes can be *idempotent* if the conditional check is on the same attribute that is being updated. DynamoDB performs a given write request only if certain attribute values in the item match what you expect them to be at the time of the request.
- Expressions
 - To get only a few attributes of an item, use a **projection expression**.
 - An **expression attribute name** is a placeholder that you use in an expression, as an alternative to an actual attribute name. An expression attribute name must begin with a #, and be followed by one or more alphanumeric characters.
 - **Expression attribute values** are substitutes for the actual values that you want to compare — values that you might not know until runtime. An expression attribute value must begin with a ;, and be followed by one or more alphanumeric characters.
 - For *PutItem*, *UpdateItem* and *DeleteItem* operations, you can specify a **condition expression** to determine which items should be modified. If the condition expression evaluates to true, the operation succeeds; otherwise, the operation fails.
 - An **update expression** specifies how *UpdateItem* will modify the attributes of an item—for example, setting a scalar value, or removing elements from a list or a map.



Time To Live (TTL)

- Allows you to define when items in a table expire so that they can be automatically deleted from the database.

DynamoDB Queries

- The *Query* operation finds items based on primary key values. You can query any table or secondary index that has a composite primary key (a partition key and a sort key).
- A key condition expression is a search criteria that determines the items to be read from the table or index.
- You must specify the partition key name and value as an equality condition.
- You can optionally provide a second condition for the sort key. The sort key condition must use one of the following comparison operators: =, <, <=, >, >=, BETWEEN, AND
- A single *Query* operation can retrieve a maximum of 1 MB of data.
- For further refining of Query results, you can optionally provide a **filter expression**, to determine which items within the Query results should be returned to you. All of the other results are discarded.
- The Query operation allows you to limit the number of items that it returns in the result by setting the **Limit** parameter to the maximum number of items that you want.
- DynamoDB paginates the results from Query operations, where Query results are divided into "pages" of data that are 1 MB in size (or less).
- **ScannedCount** is the number of items that matched the key condition expression, before a filter expression (if present) was applied.
- **Count** is the number of items that remain, after a filter expression (if present) was applied.

DynamoDB Scans

- A *Scan* operation reads every item in a table or a secondary index. By default, a Scan operation returns all of the data attributes for every item in the table or index.
- Scan always returns a result set. If no matching items are found, the result set will be empty.
- A single Scan request can retrieve a maximum of 1 MB of data.
- You can optionally provide a filter expression.
- You can limit the number of items that is returned in the result.
- DynamoDB paginates the results from Scan operations.
- ScannedCount is the number of items evaluated, before any ScanFilter is applied.
- Count is the number of items that remain, after a filter expression (if present) was applied.
- A Scan operation performs eventually consistent reads, by default.
- By default, the Scan operation processes data sequentially.

On-Demand Backup and Restore

- You can use IAM to restrict DynamoDB backup and restore actions for some resources.



- All backup and restore actions are captured and recorded in AWS CloudTrail.
- Backups
 - Each time you create an on-demand backup, the entire table data is backed up.
 - All backups and restores in DynamoDB work without consuming any provisioned throughput on the table.
 - DynamoDB backups do not guarantee causal consistency across items; however, the skew between updates in a backup is usually much less than a second.
 - You can restore backups as new DynamoDB tables in other regions.
 - Included in the backup are:
 - Database data
 - Global secondary indexes
 - Local secondary indexes
 - Streams
 - Provisioned read and write capacity
 - While a backup is in progress, you can't do the following:
 - Pause or cancel the backup operation.
 - Delete the source table of the backup.
 - Disable backups on a table if a backup for that table is in progress.
- Restore
 - You cannot overwrite an existing table during a restore operation.
 - You restore backups to a new table.
 - For tables with even data distribution across your primary keys, the restore time is proportional to the largest single partition by item count and not the overall table size.
 - If your source table contains data with significant skew, the time to restore may increase.

DynamoDB Transactions

- Amazon DynamoDB transactions simplify the developer experience of making coordinated, all-or-nothing changes to multiple items both within and across tables.
- Transactions provide atomicity, consistency, isolation, and durability (ACID) in DynamoDB, helping you to maintain data correctness in your applications.
- You can group multiple Put, Update, Delete, and ConditionCheck actions. You can then submit the actions as a single TransactWriteItems operation that either succeeds or fails as a unit.
- You can group and submit multiple Get actions as a single TransactGetItems operation.
- Amazon DynamoDB supports up to 25 unique items and 4 MB of data per transactional request.

Global Tables

- Global tables provide a solution for deploying a multi-region, multi-master database, without having to build and maintain your own replication solution.
- You specify the AWS regions where you want the table to be available. DynamoDB performs all tasks to create identical tables in these regions, and propagate ongoing data changes to all of them.



- Replica Table (Replica, for short)
 - A single DynamoDB table that functions as a part of a global table.
 - Each replica stores the same set of data items.
 - Any given global table can only have one replica table per region.
 - You can add new or delete replicas from global tables.
- To ensure eventual consistency, DynamoDB global tables use a “*last writer wins*” reconciliation between concurrent updates, where DynamoDB makes a best effort to determine the last writer.
- If a single AWS region becomes isolated or degraded, your application can redirect to a different region and perform reads and writes against a different replica table. DynamoDB also keeps track of any writes that have been performed, but have not yet been propagated to all of the replica tables.
- Requirements for adding a new replica table
 - The table must have the same partition key as all of the other replicas.
 - The table must have the same write capacity management settings specified.
 - The table must have the same name as all of the other replicas.
 - The table must have DynamoDB Streams enabled, with the stream containing both the new and the old images of the item.
 - None of the replica tables in the global table can contain any data.
- If global secondary indexes are specified, then the following conditions must also be met:
 - The global secondary indexes must have the same name.
 - The global secondary indexes must have the same partition key and sort key (if present).

Security

- Encryption
 - Encrypts your data at rest using an AWS Key Management Service (AWS KMS) managed encryption key for DynamoDB.
 - Encryption at rest can be enabled only when you are creating a new DynamoDB table.
 - After encryption at rest is enabled, it can't be disabled.
 - Uses AES-256 encryption.
 - The following are encrypted:
 - DynamoDB base tables
 - Local secondary indexes
 - Global secondary indexes
 - Authentication and Access Control
 - Access to DynamoDB requires credentials.
 - Aside from valid credentials, you also need to have permissions to create or access DynamoDB resources.
 - Types of Identities
 - **AWS account root user**
 - **IAM user**
 - **IAM role**



- You can create indexes and streams only in the context of an existing DynamoDB table, referred to as *subresources*.
- Resources and subresources have unique Amazon Resource Names (**ARNs**) associated with them.
- A *permissions policy* describes who has access to what.
 - Identity-based Policies
 - Attach a permissions policy to a user or a group in your account
 - Attach a permissions policy to a role (grant cross-account permissions)
 - Policy Elements
 - Resource - use an ARN to identify the resource that the policy applies to.
 - Action - use action keywords to identify resource operations that you want to allow or deny.
 - Effect - specify the effect, either allow or deny, when the user requests the specific action.
 - Principal - the user that the policy is attached to is the implicit principal.
- Web Identity Federation - Customers can sign in to an identity provider and then obtain temporary security credentials from AWS Security Token Service (AWS STS).

Monitoring

- Automated tools:
 - **Amazon CloudWatch Alarms** – Watch a single metric over a time period that you specify, and perform one or more actions based on the value of the metric relative to a given threshold over a number of time periods.
 - **Amazon CloudWatch Logs** – Monitor, store, and access your log files from AWS CloudTrail or other sources.
 - **AWS CloudTrail Log Monitoring** – Share log files between accounts, monitor CloudTrail log files in real time by sending them to CloudWatch Logs, write log processing applications in Java, and validate that your log files have not changed after delivery by CloudTrail.
- Using the information collected by CloudTrail, you can determine the request that was made to DynamoDB, the IP address from which the request was made, who made the request, when it was made, and additional details.

DynamoDB Accelerator (DAX)

- DAX is a fully managed, highly available, in-memory cache for DynamoDB.
- **DynamoDB Accelerator (DAX)** delivers microsecond response times for accessing eventually consistent data.
- It requires only minimal functional changes to use DAX with an existing application since it is API-compatible with DynamoDB.
- For read-heavy or bursty workloads, DAX provides increased throughput and potential cost savings by reducing the need to overprovision read capacity units.



- DAX lets you scale on-demand.
- DAX is fully managed. You no longer need to do hardware or software provisioning, setup and configuration, software patching, operating a reliable, distributed cache cluster, or replicating data over multiple instances as you scale.
- DAX is not recommended if you need strongly consistent reads
- DAX is useful for read-intensive workloads, but not write-intensive ones.
- DAX supports server-side encryption but not TLS.
- Use Cases
 - Applications that require the fastest possible response time for reads.
 - Applications that read a small number of items more frequently than others. For example, limited-time on-sale items in an ecommerce store.
 - Applications that are read-intensive, but are also cost-sensitive. Offload read activity to a DAX cluster and reduce the number of read capacity units that you need to purchase for your DynamoDB tables.
 - Applications that require repeated reads against a large set of data. This will avoid eating up all your DynamoDB resources which are needed by other applications.
- To achieve high availability for your application, provision your DAX cluster with at least three nodes, then place the nodes in multiple Availability Zones within a Region.
- There are two options available for scaling a DAX cluster:
 - **Horizontal scaling**, where you add read replicas to the cluster. A single DAX cluster supports up to 10 read replicas, and you can add or remove replicas while the cluster is running.
 - **Vertical scaling**, where you select different node types. Larger nodes enable the cluster to store more data in memory, reducing cache misses and improving overall application performance. You can't modify the node types on a running DAX cluster. Instead, you must create a new cluster with the desired node type.

Best Practices

- Know the Differences Between Relational Data Design and NoSQL

Relational database systems (RDBMS)	NoSQL database
In RDBMS, data can be queried flexibly, but queries are relatively expensive and don't scale well in high-traffic situations.	In a NoSQL database such as DynamoDB, data can be queried efficiently in a limited number of ways, outside of which queries can be expensive and slow.



In RDBMS, you design for flexibility without worrying about implementation details or performance. Query optimization generally doesn't affect schema design, but normalization is very important.	In DynamoDB, you design your schema specifically to make the most common and important queries as fast and as inexpensive as possible. Your data structures are tailored to the specific requirements of your business use cases.
For an RDBMS, you can go ahead and create a normalized data model without thinking about access patterns. You can then extend it later when new questions and query requirements arise. You can organize each type of data into its own table.	For DynamoDB, by contrast, you shouldn't start designing your schema until you know the questions it will need to answer. Understanding the business problems and the application use cases up front is essential. You should maintain as few tables as possible in a DynamoDB application. Most well designed applications require only one table.
	It is important to understand three fundamental properties of your application's access patterns: 1. Data size: Knowing how much data will be stored and requested at one time will help determine the most effective way to partition the data. 2. Data shape: Instead of reshaping data when a query is processed, a NoSQL database organizes data so that its shape in the database corresponds with what will be queried. 3. Data velocity: DynamoDB scales by increasing the number of physical partitions that are available to process queries, and by efficiently distributing data across those partitions. Knowing in advance what the peak query loads might be helps determine how to partition data to best use I/O capacity.

- Design and Use Partition Keys Effectively
 - DynamoDB provides some flexibility in your per-partition throughput provisioning by providing **burst capacity**.



- To better accommodate uneven access patterns, **DynamoDB adaptive capacity** enables your application to continue reading and writing to 'hot' partitions without being throttled, by automatically increasing throughput capacity for partitions that receive more traffic.
- Amazon DynamoDB now applies adaptive capacity in real time in response to changing application traffic patterns, which helps you maintain uninterrupted performance indefinitely, even for imbalanced workloads. In addition, instant adaptive capacity helps you provision read and write throughput more efficiently instead of overprovisioning to accommodate uneven data access patterns. Instant adaptive capacity is on by default at no additional cost for all DynamoDB tables and global secondary indexes.
- The optimal usage of a table's provisioned throughput depends not only on the workload patterns of individual items, but also on the partition-key design. In general, you will use your provisioned throughput more efficiently as the ratio of partition key values accessed to the total number of partition key values increases.
- Structure the primary key elements to avoid one heavily requested partition key value that slows overall performance.
- Distribute loads more evenly across a partition key space by adding a random number to the end of the partition key values. Then you randomize the writes across the larger space.
- A randomizing strategy can greatly improve write throughput, but it's difficult to read a specific item because you don't know which suffix value was used when writing the item. Instead of using a random number to distribute the items among partitions, use a number that you can calculate based upon something that you want to query on.
- Distribute write activity efficiently during data upload by using the sort key to load items from each partition key value, keeping more DynamoDB servers busy simultaneously and improving your throughput performance.
- Use Sort Keys to Organize Data
 - Well-designed sort keys gather related information together in one place where it can be queried efficiently.
 - Composite sort keys let you define hierarchical (one-to-many) relationships in your data that you can query at any level of the hierarchy.
- Use indexes efficiently by keeping the number of indexes to a minimum and avoid indexing tables that experience heavy write activity.
- Choose Projections Carefully.
- Optimize Frequent Queries to Avoid Fetches.
- Be Aware of Item-Collection Size Limits When Creating Local Secondary Indexes.
- For Querying and Scanning Data
 - Performance considerations for scans
 - Avoiding sudden spikes in read activity
 - Taking advantage of parallel scans

Pricing



- DynamoDB charges per GB of disk space that your table consumes. The first 25 GB consumed per month is free.
- DynamoDB charges for Provisioned Throughput --- WCU and RCU, Reserved Capacity and Data Transfer Out.
- You should round up to the nearest KB when estimating how many capacity units to provision.
- There are additional charges for DAX, Global Tables, On-demand Backups (per GB), Continuous backups and point-in-time recovery (per GB), Table Restorations (per GB), and Streams (read request units).

References:

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/Introduction.html?shortFooter=true>
<https://aws.amazon.com/dynamodb/faqs/>

AWS Fargate

- A serverless compute engine for containers that works with both Amazon Elastic Container Service (ECS) and Amazon Elastic Kubernetes Service (EKS).
- With Fargate, no manual provisioning, patching, cluster capacity management, or any infrastructure management required.
- **Use Case**
 - Launching containers without having to provision or manage EC2 instances.
 - If you want a managed service for container cluster management.
- **Configurations**
 - Amazon ECS task definitions for Fargate require that you specify CPU and memory at the task level (task definition).
 - Amazon ECS task definitions for Fargate support the limits parameter to define the resource limits to set for a container.
 - Amazon ECS task definitions for Fargate support the awslogs, splunk, firelens, and fluentd log drivers for the log configuration.
 - When provisioned, each Fargate task receives the following storage:
 - 10 GB of Docker layer storage
 - An additional 4 GB for volume mounts.
 - Task storage is ephemeral.
 - If you have a service with running tasks and want to update their platform version, you can update your service, specify a new platform version, and choose Force new deployment. Your tasks are redeployed with the **latest** platform version.
 - If your service is scaled up without updating the platform version, those tasks receive the platform version that was specified on the service's current deployment.
- **Network**



- Amazon ECS task definitions for Fargate require that the network mode is set to awsvpc. The awsvpc network mode provides each task with its own elastic network interface.
- **Compliance**
 - PCI DSS Level 1, ISO 9001, ISO 27001, ISO 27017, ISO 27018, SOC 1, SOC 2, SOC 3, and HIPAA
 - AWS Fargate is not yet available in AWS GovCloud.
- **Pricing**
 - You pay for the amount of vCPU and memory resources consumed by your containerized applications.

References:

<https://aws.amazon.com/fargate/>

<https://aws.amazon.com/fargate/faqs/>

https://docs.aws.amazon.com/AmazonECS/latest/developerguide/AWS_Fargate.html

AWS WAF

- A web application firewall that helps protect web applications from attacks by allowing you to configure rules that **allow, block, or monitor (count) web requests** based on conditions that you define.
- These conditions include:
 - IP addresses
 - HTTP headers
 - HTTP body
 - URI strings
 - SQL injection
 - cross-site scripting.

Features

- WAF lets you create rules to filter web traffic based on conditions that include IP addresses, HTTP headers and body, or custom URIs.
- You can also create rules that block common web exploits like SQL injection and cross site scripting.
- For application layer attacks, you can use WAF to respond to incidents. You can set up proactive rules like *Rate Based Blacklisting* to automatically block bad traffic, or respond immediately to incidents as they happen.
- WAF provides real-time metrics and captures raw requests that include details about IP addresses, geo locations, URIs, User-Agent and Referers.
- **AWS WAF Security Automations** is a solution that automatically deploys a single web access control list (web ACL) with a set of AWS WAF rules designed to filter common web-based attacks. The solution supports log analysis using Amazon Athena and AWS WAF full logs.

Conditions, Rules, and Web ACLs



- You define your conditions, combine your conditions into rules, and combine the rules into a web ACL.
- **Conditions** define the basic characteristics that you want WAF to watch for in web requests.
- You combine conditions into **rules** to precisely target the requests that you want to allow, block, or count. WAF provides two types of rules:
 - **Regular rules** - use only conditions to target specific requests.
 - **Rate-based rules** - are similar to regular rules, with a rate limit. Rate-based rules count the requests that arrive from a specified IP address every five minutes. The rule can trigger an action if the number of requests exceed the rate limit.
- **WAF Managed Rules** are an easy way to deploy pre-configured rules to protect your applications common threats like application vulnerabilities. All Managed Rules are automatically updated by AWS Marketplace security Sellers.
- After you combine your conditions into rules, you combine the rules into a **web ACL**. This is where you define an action for each rule—allow, block, or count—and a default action, which determines whether to allow or block a request that doesn't match all the conditions in any of the rules in the web ACL.

Pricing

- WAF charges based on the number of web access control lists (web ACLs) that you create, the number of rules that you add per web ACL, and the number of web requests that you receive.

References:

<https://docs.aws.amazon.com/waf/latest/developerguide>
<https://aws.amazon.com/waf/features/>
<https://aws.amazon.com/waf/pricing/>
<https://aws.amazon.com/waf/faqs/>

AWS Shield

- A managed Distributed Denial of Service (DDoS) protection service that safeguards applications running on AWS.

Shield Tiers and Features

- **Standard**
 - All AWS customers benefit from the automatic protections of Shield Standard.
 - Shield Standard provides always-on network flow monitoring which inspects incoming traffic to AWS and detect malicious traffic in real-time.
 - Uses several techniques like deterministic packet filtering, and priority based traffic shaping to automatically mitigate attacks without impact to your applications.



- When you use Shield Standard with CloudFront and Route 53, you receive comprehensive availability protection against all known infrastructure attacks.
- You can also view all the events detected and mitigated by AWS Shield in your account.
- **Advanced**
 - Shield Advanced provides enhanced detection, inspecting network flows and also monitoring application layer traffic to your Elastic IP address, Elastic Load Balancing, CloudFront, or Route 53 resources.
 - It handles the majority of DDoS protection and mitigation responsibilities for **layer 3**, **layer 4**, and **layer 7** attacks.
 - You have 24x7 access to the AWS DDoS Response Team. To contact the DDoS Response Team, customers will need the Enterprise, Business Support+, or Unified Operations levels of AWS Premium Support.
 - It automatically provides additional mitigation capacity to protect against larger DDoS attacks. The DDoS Response Team also applies manual mitigations for more complex and sophisticated DDoS attacks.
 - It gives you complete visibility into DDoS attacks with near real-time notification via CloudWatch and detailed diagnostics on the “AWS WAF and AWS Shield” Management Console.
 - Shield Advanced comes with “DDoS cost protection”, a safeguard from scaling charges as a result of a DDoS attack that cause usage spikes on your AWS services. It does so by providing service credits for charges due to usage spikes.
 - It is available globally on all CloudFront and Route 53 edge locations.
 - With Shield Advanced you will be able to see the history of all incidents in the trailing 13 months.

Pricing

- **Shield Standard** provides protection at no additional charge.
- **Shield Advanced**, however, is a paid service. It requires a 1-year subscription commitment and charges a monthly fee, plus a usage fee based on data transfer out from CloudFront, ELB, EC2, and AWS Global Accelerator.

References:

<https://aws.amazon.com/shield/features/>
<https://aws.amazon.com/shield/pricing/>
<https://aws.amazon.com/shield/faqs/>



AWS Developer Services

AWS has a lot of tools and services that can help you accelerate your application development – enabling you to create feature-rich apps in no time! These services can help you expedite the development of your web and mobile apps; easily add a user authentication capability or social logins to your applications; launch production-ready REST and GraphQL APIs in a matter of days instead of weeks or months, and deploy, manage, and connect to a data store easily.

We will cover the related development services in AWS for the AWS Certified Solutions Architect exam in this section.

AWS Amplify

AWS Amplify is one of the many development services in AWS that helps you build extensible, full-stack web and mobile apps faster. It allows you to easily start your application development and automatically scale your resources with less management overhead. In other words, AWS Amplify deploys your application in a serverless architecture without any manual intervention on your part. Plus, it also gives you the option to integrate Machine Learning capabilities into your applications!

Put simply, AWS Amplify is a set of purpose-built tools and features. It consists of the Amplify Studio, Amplify Libraries, Amplify CLI, Amplify Hosting, and many more. Let's quickly cover its different modules one by one.

Amplify Studio is a visual development environment that simplifies the development of your web and mobile apps. You can easily build your frontend UI with its set of ready-to-use UI components, which you can directly connect to your app backend. Since your UI components are already pre-built, the number of your boilerplate code will be reduced, allowing you more time to focus on implementing the application business logic.

You can define your data models, implement user authentication, and add file storage using the Amplify Studio without any backend expertise. In addition, you can import Figma prototypes built by your application designers into the Amplify Studio for more seamless collaboration.

The AWS Amplify Libraries are open-source JavaScript libraries in the Amplify Framework. These are specifically designed to aid you in building AWS-powered mobile and web apps. The Amplify JavaScript libraries are supported in React, React Native, Angular, Ionic, Vue, and different web and mobile frameworks.

The Amplify Command Line Interface is a CLI toolchain that you can use to configure and maintain your app backend straight from your local desktop. The Amplify CLI has an interactive workflow and intuitive use cases that you can leverage, such as authentication, storage, and API. It allows you to test your features locally and deploy your app to multiple environments. This tool provides infrastructure-as-code templates which are automatically loaded to CloudFormation or AWS SAM. With the Amplify CLI, you can have a much more effective team collaboration and easy integration with Amplify's CI/CD workflow.



It also has a feature called Amplify Hosting, which allows you to host secure, reliable, fast web apps or websites via the AWS content delivery network. Under the hood, it deploys your applications to Amazon CloudFront's content delivery network, which comes with hundreds of points of presence globally. Amplify Hosting also allows you to set up your own custom domain for your applications and add custom alarms that send notifications if a certain metric exceeds a threshold that you specify.

These are some of the tools and features available in AWS Amplify that you can leverage with. Aside from Amplify Studio, Amplify Libraries, Amplify CLI, and Amplify Hosting, there are many other features and tools in AWS Amplify that you can try out, and AWS is adding more features soon to fast-track your software development experience!

AWS Device Farm

AWS Device Farm is an app testing service for testing and interacting with your Android, iOS, and web apps on real, physical phones and tablets. Instead of doing the manual testing yourself, you can just use this service so you can do a wide range of tests without having to provision and manage any testing infrastructure.

AWS Device Farm supports both native and hybrid Android, iOS, and progressive web apps. Cross-platform frameworks are also supported, such as PhoneGap, Titanium, Xamarin, Unity, and others.

This service is also capable of running Selenium tests on different desktop browsers and browser versions that are hosted in the AWS Cloud. AWS Device Farm can even execute your test suite on your own local machine and interact with browsers hosted on AWS Device Farm through the Selenium API.

This accelerates your mobile app development workflow by executing tests on multiple devices without the administrative overhead of maintaining a test environment. With AWS Device Farm, you can do integration tests that check and detect any customer issues before they are even released in production.

Amazon Managed Grafana

Amazon Managed Grafana is a fully managed service for Grafana. Grafana is an open-source analytics platform that is commonly used to query, visualize, observe, and make use of your system data that are gathered from multiple sources. When you hear the phrase "Amazon Managed", that means AWS is managing the underlying infrastructure required to run an open-source program or a particular tool. For Amazon Managed Grafana, AWS is the one that provisions and manages the required resources to run your Grafana dashboards, along with its other dependencies.



Let's take a good look at what Grafana really is. What you see here is an actual Grafana platform. It has tables, graphs, time series data, maps, statistics, and a range of other useful metrics. On its sidebar menu, you'll see the Dashboards, Explorer, Alert, Grafana Machine Learning, Kubernetes Monitoring, Synthetic Monitoring, and so much more. Grafana can collect metrics from different data sources so you can have a single place for all your data. In this way, all the metrics from your databases, servers, storage systems, APIs, custom data, logs, and other sources can be aggregated in one location. This is also the case in Amazon Managed Grafana.

In AWS, Amazon Managed Grafana can collect system metrics from multiple data sources in your observability stack, such as Amazon Managed Service for Prometheus, Amazon CloudWatch, and Amazon OpenSearch Service. System alerts can also be automated by using different notification services in AWS. You can also integrate this with third-party vendors like Datadog, Splunk et cetera. In addition, you can set up your own self-managed data source like InfluxDB and integrate it with your Grafana workspace. You also don't have to worry about the infrastructure required to run your Grafana visualizations and dashboards since the necessary resources are all provisioned and managed by AWS itself.

Amazon Managed Grafana is somewhat similar to Amazon CloudWatch Dashboard since both of them have a dashboard where you can see your system metrics. However, Grafana has a much better user interface and can integrate easier with a range of other systems and data providers.

Amazon Managed Service for Prometheus

Amazon Managed Service for Prometheus is a fully managed service for the open-source monitoring tool called Prometheus. This is commonly used for monitoring modern cloud-native applications and Kubernetes clusters. Prometheus can enable you to securely ingest, store, and query metrics from different container environments.

In Amazon Managed Service for Prometheus, the resources required to run the open-source Prometheus tool are all provisioned and managed by the AWS team themselves. Scaling the underlying architecture is also handled by Amazon. You have the option to collect system metrics from your container clusters running in AWS, running on-premises, or even both.

Amazon Managed Service for Prometheus allows you to use the open-source Prometheus query language or PromQL. This query language helps you to monitor the performance of your containerized workloads that are running on the AWS Cloud or on-site. It can also scale automatically as your workloads increase or shrink and uses AWS security services to enable fast and secure access to data.

Amazon Managed Service for Prometheus has a feature called Alert Manager that handles all alerts that are sent from your workspace. These alerts are grouped and routed to downstream receivers that you specify. Duplicate alerts and messages can also be deduplicated by Alert Manager. Under the hood, it uses the Amazon Simple Notification Service as a receiver and routes messages to different SNS topics in the same account.



In this service, you create a workspace for your Prometheus metrics. A workspace isolates access control to a particular application suite for ingestion, storage, and querying of your metrics. The container metrics that are available in your Prometheus workspace can also be collected by the Amazon Managed Grafana service.

AWS Machine Learning Services

Machine learning is a branch of artificial intelligence that is all about imitating how humans learn. Just like humans, a machine can learn by watching, hearing, reading, and analyzing the things in its environment. This machine can be in the form of a simple computer program, an API, or an entire enterprise system. Once a machine consumes enough information, it can provide a range of helpful information and useful results that can save you time, effort, and even money.

Some people may find this topic to be quite difficult to understand, especially with all the complex math and algorithms it entails. To make it simple, let's start with a couple of real-world machine learning applications that you might be using already.

Most of us have a subscription to online streaming services like Netflix, Amazon Prime Video, Disney Plus, or HBO. In these streaming services, you might notice that they can recommend a particular movie or TV series based on shows that you have watched in the past. The recommendation engine will analyze your viewing history and predict the genre or type of shows you like. So if you are watching a lot of horror movies, you will be presented with different horror shows since this is the genre that you are always watching.

Let's look at another use case. Do you know that if you take a photo on your iPhone, it will automatically detect any text in the image? In this way, you can easily copy text and not manually type them on your phone. You can even try it right now! Take a picture of something that has a label, a word, or a number on it. Open that picture, then double-tap any text present in the image, and you will be able to copy them right away. This iPhone feature is called Live Text, which is a form of Optical Character Recognition that uses machine learning to transcribe text from images in your Photo Library.

Here's another example: if you're living in the US, you might have shopped at Amazon Go which is a convenience store with no checkout! Amazon Go uses computer vision and machine learning to automatically track all the products you grabbed from the store and bill you online without having to wait in line to check out your goods! This computer vision technology is also what powers self-driving cars, facial recognition systems, and many others. There is no doubt that we are already using Machine Learning in almost every moment of our lives.

In this section, we will learn about the various Machine Learning services that are available in Amazon Web Services. The primary machine learning platform in AWS is called Amazon SageMaker AI, which is followed by a lot of other ML services. Some services in AWS have machine learning features as well, like Amazon Aurora



Machine Learning, Redshift ML, Deep Learning AMLs, and so much more. Take note that we won't cover them here as we will only focus on the dedicated ML services that are relevant to your upcoming AWS exam.

It is easy to get overwhelmed by the sheer number of services in AWS, so we will divide this lecture into several sections. The AWS Machine Learning services can be classified into these use cases: Computer Vision, Language AI, Automated Data Extraction and Analysis, Customer Experience Improvement, Business Metrics, and DevOps.

For Computer Vision, we have:

- Amazon Rekognition
- Amazon Lookout for Vision
- AWS Panorama

For Automated data extraction and analysis, AWS has:

- Amazon Textract
- Amazon Augmented AI
- Amazon Comprehend

For Language AI, you have:

- Amazon Lex
- Amazon Transcribe
- Amazon Polly

For Customer Experience improvement, you can use the following services:

- Amazon Kendra
- Amazon Personalize
- Amazon Translate

For Business metrics, you can use:

- Amazon Fraud Detector

And lastly, for DevOps, we have:

- Amazon DevOps Guru
- Amazon CodeGuru Reviewer
- Amazon CodeGuru Profiler
- Amazon CodeWhisperer



Let's discuss each of these services one by one.

Amazon SageMaker AI

First off, let's talk about Amazon SageMaker AI. Think of this as a full-fledged machine learning platform in AWS with tons of services, features, and components. Amazon SageMaker AI is not just a simple ML service but a fully managed cloud platform with lots of modules that you can use. With this, you can build, train, and deploy ML models for any use case with fully managed infrastructure, tools, and workflows. SageMaker removes the manual tasks from each step of the ML process to make it easier for you to develop high-quality models. This platform has so many modules that you choose from, namely: Amazon SageMaker AI Canvas, SageMaker Studio Lab, SageMaker Data Wrangler, SageMaker Autopilot, SageMaker JumpStart, SageMaker Clarify, and so much more!

Amazon Rekognition

Amazon Rekognition provides pre-trained and customizable computer vision capabilities to extract information and insights from your images and videos. Just as its name implies, it can recognize certain objects, faces, texts, scenes, labels, and other attributes from your media files or streaming videos. This service is perfect for facial recognition, where it can detect the face of a particular person or a well-known celebrity. It can also determine if someone is wearing a piece of Personal Protective Equipment like a mask, a helmet, or gloves. If you upload an image of you holding a guitar while sitting on your sofa, Amazon Rekognition can detect your face, your guitar, and the sofa where you sit. It also has a feature called Amazon Rekognition Custom Labels. With this, you can easily build a machine learning model to classify custom components or products from your dataset.

Amazon Lookout for Vision

Amazon Lookout is a suite of services that is comprised of Amazon Lookout for Equipment, and Amazon Lookout for Vision. The last one uses computer vision to detect defects on industrial products at scale. Amazon Lookout for Vision is primarily used in factories and manufacturing lines to quickly and accurately identify defects in each product. The dataset can be in a form of product images that are stored in an Amazon S3 bucket. You can provide a couple of baseline images to Amazon Lookout for Vision containing defect-free products, and this service will be able to automatically build a model for you within a few hours. From there, it can automatically detect anomalies in your product like dents, cracks, and scratches.



Amazon Textract

The name of this service is a combination of the words "text" and "extract". This will give you a hint that it is used to extract texts from scanned documents, notes, and images. Amazon Textract is a service that uses optical character recognition to automatically extract text from scanned files like PDFs, Word documents, hand-written notes, receipts, passports, IDs, and many others. What makes this service great is its capability to generate the results into a table form or a CSV file. It also has a query feature that allows you to extract a particular field using natural language questions. So if you upload your driver's license to Amazon Textract, you can submit a query like "What's the first name?" or "What's the driver's ID?" and you'll get the value you asked for. You can also batch upload your documents to S3 and automate the text analysis process.

Amazon Augmented AI

Amazon Augmented AI or A2I is a service that provides human review workflows for common machine learning use cases. A human review literally means that a human being will review a certain output that your machine learning model generated before it can proceed to the next step of the workflow. This service augments your AI to ensure the accuracy of prediction results and helps provide continuous improvements to your machine learning model. You can directly integrate your workflows from Amazon Rekognition or Amazon Textract to Amazon Augmented AI. For example, you can do a human review of the key-value pairs that are extracted by Amazon Textract or implement image moderation by doing a human review of unsafe content, such as explicit adult or violent content from Amazon Rekognition. It is also possible to run a human review with a custom machine learning workflow of your choice.

Amazon Comprehend

Amazon Comprehend is a natural language processing service in AWS that can find insights and relationships in a text. It performs text analytics that can automatically extract key phrases, sentiment, language, syntax, topics, and even Personally Identifiable Information or PII from unstructured data. In essence, Amazon Comprehend is a service that comprehends or understands the information written in your text documents. This is different from Amazon Textract since Amazon Comprehend cannot read text from scanned documents. You need to have raw text data first in order to use Amazon Comprehend.



Amazon Lex

Amazon Lex is a machine learning service that allows you to develop chatbots. You can build Voice-based or Text-based chatbots with Amazon Lex easily. This is helpful if your company needs a self-service bot or a virtual agent for your conversational Interactive Voice Response (IVR) system, corporate website, or other customer-facing application. Amazon Lex can significantly reduce the costs of companies in maintaining its contact center.

Amazon Transcribe

Amazon Transcribe is simply a speech-to-text transcription service. The word transcribe means to make a written record of a speech, a phone call, or any spoken language, and this is exactly what Amazon Transcribe does. It is also helpful in contact centers as it can generate call transcripts and provide conversation insights to help improve customer experience and agent productivity. Amazon Transcribe also offers real-time transcription – where you can just talk to its endpoint, and it will immediately generate transcripts of your speech.

Amazon Polly

The other service relating to Language AI is called Amazon Polly. Essentially, Amazon Polly is the exact opposite of the Amazon Transcribe service. Instead of turning speech into text, it converts text into speech! If you input a text into Amazon Polly, it will generate a lifelike speech in different voices that you specify. So, for example, you typed: “Beautiful Philippine Islands” in the Amazon Polly console. You can hear the phrase: “Beautiful Philippine Islands” in a male voice, a female voice, a kid's voice, or in any voice that you prefer. You can also customize the pronunciation of specific words and phrases by uploading your own lexicon files. A lexicon is simply a vocabulary of a particular language, and this is usually used if you have a non-English text that you want to turn into speech.

Amazon Kendra

Amazon Kendra is an intelligent search service in AWS. It is not just a typical search service that simply returns a match to your query from a single data source. Amazon Kendra can search from multiple data sources that can be structured or unstructured, then intelligently analyze the content before it sends a result. This service supports natural language processing, so you can ask questions using a language that you use in your everyday life. For instance, you can ask Amazon Kendra, “Who is the founder of the EdTech startup: Tutorials Dojo?” and it will search all of the documents in your S3 bucket, Amazon FSx file systems, Amazon RDS databases, Github repository, Jira, Slack, Sharepoint and other data sources for the answer. Again, it can search for information from a wide range of sources and not just from a traditional SQL database, then uses machine learning to provide context to your search results.



Amazon Personalize

Amazon Personalize is a service that provides personalized recommendations to your customers based on their past activity and behavior. It's just like the recommendation feature in Amazon Prime or Netflix, where new movies are automatically recommended for you based on your viewing history. If you watched a lot of Sci-Fi movies on their online streaming platform, they will automatically recommend more Sci-Fi shows on your profile. This is definitely a customer experience improvement since personalizing the user's content tends to convert more because they align with what the customers actually do and buy.

Amazon Translate

Amazon Translate is a real-time translation service in AWS. It works pretty much like Google Translate, where you input text in one language, and the service will translate it to a language that you choose. You can also create your own custom terminology. This allows you to customize the output of Amazon Translate based on a company-specific and domain-specific vocabulary. For example, I can set the acronym TD as "Tutorials Dojo" in English. The Amazon Translate service can accept input with my custom vocabulary and include it in the translation. In this case, I can enter the Tagalog phrase "Magandang Umaga TD" and Amazon Translate will return "Good morning Tutorials Dojo" as an output. The Filipino phrase "magandang umaga" means "good morning" in English, while "TD" is the custom term for "Tutorials Dojo" which we configured in Amazon Translate. You can also enable the Formality option, which controls whether the translation output uses a formal tone or not. The translation can also mask profane words or phrases, which is a very useful feature for customer-facing applications.

Amazon Fraud Detector

Amazon Fraud Detector is yet another machine learning service that can automate fraud detection, just as its name suggests. It can identify potential fraudulent activity, fake reviews and spam account creation in near-real-time. For instance, your website recently got a visitor whose IP address has a history of malicious activity such as spamming, hacking attempts, and DDoS attacks. Users with exactly the same IP address are posting spam on your website repeatedly. For this situation, you can use Amazon Fraud Detector to block any visitor who uses an offending IP address, an email domain, or any other attribute that you define.

Amazon DevOps Guru

Amazon DevOps Guru detects abnormal behavior in your application or AWS resources that might cause unexpected downtimes or operational issues in the near future. It can monitor applications and AWS resources within your own account or on all accounts across your AWS Organization. It uses machine learning to identify operational defects long before they impact you and your customers. Amazon DevOps Guru can analyze your RDS databases and automatically determine an unusually high DB load that is more than three times or 5 times its normal value. It can also detect issues in your serverless stack like an extremely high number of invocations in your Lambda function that is beyond the currently provisioned concurrency or an overprovisioned write capacity on your DynamoDB tables.



Amazon CodeGuru

Amazon CodeGuru is a suite of development services in AWS. It contains different tools and features such as Amazon CodeGuru Reviewer, Amazon CodeGuru Profiler, BugBust, and many more. The primary function of Amazon CodeGuru Reviewer is to provide intelligent recommendations for improving your application performance, efficiency, and code quality. It can scan your code and detect a plethora of code defects like bad exception handling, insecure CORS policy, path traversal, hardcoded credentials, and many more. You can also integrate this with your CI/CD workflow so you can run code reviews and recommendations to improve your codebase. The other module for this service is called the Amazon CodeGuru Profiler. A profiler is basically a component that collects your CPU data and analyzes the runtime performance data from your live applications. This is helpful in identifying expensive lines of codes that inefficiently use the CPU, which causes CPU bottlenecks.

Amazon CodeWhisperer

Amazon CodeWhisperer is a coding tool that automatically generates code and functions in real-time. This tool is similar to Github CoPilot, which is an extension that you usually install in your visual studio IDE. The lines of codes are generated right from your IDE editor based on the comments that you write. For example, you can simply write a comment that outlines a specific task in plain English, such as "Upload a file to an Amazon S3 bucket with server-side encryption". Amazon CodeWhisperer will take your comment as input and generate an entire function in the programming language that you define, which can upload a file to your S3 bucket with the required encryption and many more



AWS Deployment Services

There are different ways to build, test, and deploy your applications to your development or production environments. Back in the day, we used to just copy a ZIP file, a WAR file, or the binary files in the webapps directory of our web servers. Then, we have to manually configure and restart our Apache, NGINX, and IBM WebSphere servers to reflect the changes. This process is called a manual deployment and is prone to a lot of human errors. There are a lot of moving parts that you have to do by hand, which may take several hours to complete. Worse, provisioning the required virtual web server or a dedicated database cluster will take you weeks or even months to accomplish due to a lack of automation. Good thing that nowadays, you can do your deployment in a blink of an eye through the various deployment tools available at your disposal.

AWS offers a number of services that provide deployment and management capabilities for one or more aspects of your application lifecycle. These services enable organizations to build and deliver applications faster than their traditional CI/CD workflow. You and your team don't have to spend a lot of time manually provisioning, configuring, updating, monitoring, or securing your AWS resources anymore. These laborious tasks can be programmed into code and easily automated with the different deployment services available in AWS. Some of these services use the concept of "Infrastructure as Code" or IaC. An IaC is a process of managing and provisioning your servers, databases, CDNs, and other resources through machine-readable definition files. Simply put, you only need to provide a text-based definition file that will automatically provision the required resources for your application in just one click of a button.

The advent of cloud computing enables companies to deploy all their workloads entirely in the cloud. They also have the option to run a hybrid cloud architecture in which they utilize both their physical on-premises resources and cloud services in AWS at the same time. There's even an option now to do a multi-cloud deployment where you deploy your infrastructure to AWS, Azure, Google Cloud, and other public cloud providers simultaneously. You can run both your multitier applications and Kubernetes or Docker container cluster almost anywhere – whether it be on the cloud, on-premises, on multiple public clouds, or a combination of all three.

Let's discover the different deployment services in this lesson. These services have the capability to provision, configure, deploy, scale, and monitor your cloud architecture without any manual intervention on your part. They are:

- AWS CloudFormation
- AWS Elastic Beanstalk
- AWS CodeDeploy
- Amazon ECS Anywhere
- Amazon EKS Anywhere
- AWS Systems Manager
- AWS Proton



AWS CloudFormation

AWS CloudFormation is a service that enables you to provision and manage your AWS resources using a custom code template. You can create a custom template in YAML or JSON format that defines the AWS resources that you require, like Amazon EC2 instances, Amazon FSx filesystems, Amazon Aurora databases, CloudFront distributions, or any other resource, and the AWS CloudFormation service can deploy all of these automatically for you. AWS CloudFormation also comes with a graphic tool called the AWS Infrastructure Composer. This is a drag-n-drop online tool for creating, viewing, and modifying your AWS CloudFormation templates.

CloudFormation is the primary Infrastructure as Code service in AWS. It works like any other IaC tools in the market, like Terraform, Ansible, Chef, and Puppet. The key difference, however, is the additional features that it provides, which are fully compatible with the AWS Cloud.

A CloudFormation template deploys your cloud infrastructure resources in a group called a "stack". This stack can represent your entire cloud architecture or just its subset. If you have a large multi-tier architecture, you can create multiple templates to represent the different tiers of your enterprise application suite. You can create a CloudFormation template for your presentation layer stack, another template for your application layer stack, and one more for your data layer stack.

You can bundle your multiple stacks together into something called a nested stack. In CloudFormation, you can have a root stack with a hierarchy of nested stacks under it. This will effectively make the modules of your infrastructure code to be loosely coupled with each other, which makes the management of each individual stack much easier. Your application layer stack will only contain EC2, EKS, ECS, and other compute resources, while the data layer stack will have RDS, Aurora, DocumentDB, Amazon Neptune, Amazon Timestream, or any database services. Any change in one nested stack won't adversely affect the other stack.

Aside from provisioning your resources, Amazon CloudFormation allows you to change, modify, or scale your services that are already deployed in your AWS account. It even has a dry-run mode to check your upcoming changes before they are deployed in your cloud environment. This feature is called a "Change Set".

Essentially, a change set allows you to see how your changes might impact your running resources before finally implementing them. Say you want to change the name of your Amazon Aurora Serverless database. You can create a change set in CloudFormation that will create a new Aurora database and delete the old one. Of course, you will lose the data in the old database unless you've taken a DB snapshot for backup. The change set will show you the upcoming change so you can plan accordingly before you update your stack.

A CloudFormation stack is usually mapped in a single AWS account only, so if you are running your applications in two or more AWS accounts, you can use StackSets. A StackSet extends the capability of CloudFormation stacks by enabling you to create, update, or delete stacks across multiple accounts and AWS



Regions with a single operation. You can select an administrator account in your AWS Organization and then choose a particular CloudFormation template for your StackSet. This template will be the basis for provisioning the stacks into your selected target accounts.

The stack, change set, StackSet, and the graphical designer tool are the base features of the AWS CloudFormation service. There are different services in AWS that extend the capabilities in CloudFormation, namely the AWS Cloud Development Kit and AWS Serverless Application Model. The AWS Cloud Development Kit, or AWS CDK for short, is an open-source software development kit for Amazon Web Services. You can use this to programmatically model your AWS infrastructure using TypeScript, Python, Java, .NET, or any other programming languages that you prefer.

AWS Serverless Application Model (AWS SAM)

The AWS Serverless Application Model service, or AWS SAM, is an open-source framework that simplifies the development of your serverless applications on AWS. This is commonly used if your cloud stack is using AWS Lambda, Amazon DynamoDB, API Gateway, and other serverless services. AWS SAM can also have a SAM template that is essentially just an extension of the AWS CloudFormation template. A SAM template has some additional components that make it easier for you to work with serverless services in AWS.

Your apps can be stored in the AWS Serverless Application Repository. This repository service makes it easy for developers and companies to deploy, manage, and share their serverless applications in AWS and to the greater public. You can easily publish your serverless apps and share them with the community at large or privately within your organization. Each and every application in the AWS Serverless Application Repository is packaged with an AWS Serverless Application Model (SAM) template that defines the different AWS resources which the application uses.

AWS Elastic Beanstalk

AWS Elastic Beanstalk is a managed platform that allows you to upload your application code in AWS and provision the required cloud environment easily. You only need to upload your application package that is written in Java, .NET, PHP, Node.js, Python, Ruby, Go, or Docker, and then Elastic Beanstalk will deploy the necessary resources to run your application. You can either run a Web Server environment or a Worker environment. A Web Server environment runs a static website, a web app, or a web API that serves HTTP requests, while a worker environment, on the other hand, runs a worker application that processes long-running workloads on demand. The latter also performs tasks on a schedule that you define and can be integrated with the Amazon SQS queue. The AWS Elastic Beanstalk service also uses a configuration file like CloudFormation, to automatically deploy and configure your applications. These configuration files in Elastic Beanstalk are stored in the .ebextensions folder.



AWS CodeDeploy

AWS CodeDeploy is a fully managed deployment service that automates your application deployments in AWS. You can deploy your applications to Amazon EC2 instances, Amazon ECS clusters, AWS Lambda functions, and other computing services in AWS. You can even use this to deploy your application to the servers located on your on-premises network. This service is different from AWS CloudFormation, AWS SAM, and Elastic Beanstalk since you can only deploy your applications to existing compute resources. AWS CodeDeploy does not create AWS resources on your behalf and is intended for application deployments only.

Amazon ECS Deployment Options

Amazon ECS is a fully managed container orchestration service that supports Docker containers. This orchestration service allows you to easily run containerized applications on a managed cluster that you can control. Basically, container orchestration is an automation tool that reduces the operational effort needed to run your containerized workloads and services. Amazon ECS can automatically orchestrate or control the manual tasks of provisioning, deploying, scaling, networking, and many other tasks, so you don't have to.

Amazon ECS Anywhere is a feature of Amazon ECS that enables you to run and manage container workloads on your own on-premises infrastructure. You'll still have the same cluster management, workload scheduling, monitoring, and support features in Amazon ECS Anywhere, whether you are running your workload on-premises or in the cloud. This ubiquitous service can help you meet your compliance requirements and scale your hybrid architecture without undermining the previous investments you already have in your on-premises hardware.

When you create an Amazon ECS cluster, you can choose whether the compute resources will be deployed or launched in your own VPC, in AWS Fargate, or externally via Amazon ECS Anywhere. A cluster launched in a VPC uses Amazon EC2 instances that are orchestrated and controlled by Amazon ECS. You can use the Amazon CloudWatch Container Insights to monitor your container workloads.

If the launch type is AWS Fargate, then your cluster will be serverless, and the computing resources will be fully managed by AWS. Using AWS Fargate can significantly reduce the operating costs of running your containers. If you decide to launch your cluster externally, then your cluster will be run on your own server located on-site through the help of the Amazon ECS Anywhere service.

Amazon EKS Deployment Options

Amazon Elastic Kubernetes Service or Amazon EKS is a managed service that you can use to run Kubernetes on AWS. It's like Amazon ECS, but instead of Docker containers, this service is used for running Kubernetes clusters. Amazon EKS automates the installation, operation, and maintenance of your own Kubernetes control plane, pods, and nodes.



You can deploy your Kubernetes cluster in various ways in AWS and can include additional networking add-ons to improve your containerized architecture. A Kubernetes container can be deployed to an Amazon EKS cluster in your AWS account, to Amazon EKS on AWS Outposts, to Amazon EKS Anywhere, and through the Amazon EKS Distro. The first option allows you to launch a Kubernetes cluster using managed or self-managed Amazon EC2 nodes that you can customize and control. You can also choose to deploy your Kubernetes pods on AWS Fargate to make the cluster serverless and extremely cost-effective.

The second type is Amazon EKS on AWS Outposts which is a deployment option that uses a physical AWS Outpost rack on your on-premises network to run your Kubernetes workloads. The data plane is also located on-premises, so you can have more control compared with running it exclusively in AWS.

Using Amazon EKS Anywhere is another way to deploy your containers on-premises. It works like Amazon ECS Anywhere, which allows you to run your Kubernetes cluster entirely on your own. This means that the hardware, app deployment location, control plane, and data plane are all controlled on your own physical network. This gives you extensive control over all the components of your containerized application suite while maintaining official support from AWS.

The other deployment option that you can choose is Amazon EKS Distro. The word "distro" simply refers to the distribution of the same open-source Kubernetes software deployed by Amazon EKS in the AWS cloud. Amazon EKS Distro follows the same Kubernetes version release cycle as Amazon EKS and is provided to you as an open-source project that you can deploy on your own computer or on-site environment. It's similar to the Amazon EKS Anywhere option, except that it does not include support services offered by AWS.

AWS Proton

AWS Proton is a service that automates container and serverless deployments in AWS. It empowers your platform teams and developers to have consistent development standards and best practices. This service is very useful if you have a large number of developers in your organization. AWS Proton enables your developers to deploy container and serverless applications using pre-approved stacks that your platform team manages. It balances control and flexibility in your organization by allowing developers to innovate within the set guardrails that you implement.

It also offers a self-service portal for your developers, which contains AWS Proton templates that they can use and deploy. A Proton template contains all the information required to deploy your custom environments and services. You can create an AWS Proton Component as well, which provides flexibility to your service templates. These components in AWS Proton provide platform teams with a way to extend core infrastructure patterns and define guardrails for your developers.



AWS Audit Manager

The AWS Audit Manager is a service that you can use to continuously audit your AWS usage. It simplifies the process of doing risk and compliance assessment by utilizing a pre-built standard framework that is based on common compliance standards and AWS best practices. This service uses the data gathered from AWS Security Hub, AWS Config, AWS License Manager, and other AWS services to generate audit reports. The AWS Audit Manager can import license usage limits and the daily usage data from AWS License Manager to generate assessment reports accordingly.

Keep in mind that this is different from AWS Artifact, which only offers evidence to prove that the AWS Cloud infrastructure has met the compliance requirements. AWS Artifact does not check the AWS resources that you are actually using or even the way you use them. On the other hand, the AWS Audit Manager gathers evidence to prove that your usage of AWS services complies with the industry standards or regulations.

Amazon Inspector

Amazon Inspector is an automated security assessment service that helps improve the security and compliance of applications deployed on AWS. Amazon Inspector automatically assesses applications for vulnerabilities or deviations from best practices. After performing an assessment, Amazon Inspector produces a detailed list of security findings prioritized by level of severity. It provides an automated security assessment report that will identify unintended network access to your Amazon EC2 instances and vulnerabilities in those instances. These findings can be reviewed directly or as part of detailed assessment reports which are available via the Amazon Inspector console or API.

Amazon Detective

Amazon Detective makes it easy to analyze, investigate, and quickly identify the root cause of potential security issues or suspicious activities. Amazon Detective automatically collects log data from your AWS resources. What it basically does is to collect logs from AWS CloudTrail, Amazon VPC Flow Logs, Amazon GuardDuty findings, and other AWS services then use machine learning to analyze and conduct security investigations.

AWS Security Hub

AWS Security Hub is a service that provides a centralized and comprehensive view of the security posture of your cloud infrastructure across multiple AWS accounts. This service helps you to comply with specific security standards and best practices that your organization require. Basically, it works like a hub that collects security alerts and findings from multiple AWS services, such as Amazon GuardDuty, Amazon Inspector,



Amazon Macie, AWS Identity and Access Management (IAM) Access Analyzer, AWS Firewall Manager and other sources. You can use the AWS Security Hub to comply with the Payment Card Industry Data Security or PCI DSS Standard, Center for Internet Security or CIS Benchmarks, and many other security standards.

AWS Network Firewall

AWS Network Firewall is a managed network firewall service for your Amazon Virtual Private Clouds. This network security service is a managed network firewall that comes with intrusion prevention and detection capabilities. The AWS Network Firewall service allows you to filter traffic within the perimeter of your Amazon VPCs.

This service is commonly used in various network security use cases such as inspecting VPC-to-VPC traffic, filtering outbound traffic, securing both AWS Direct Connect connection and VPN traffic as well as filtering the Internet traffic. AWS Network Firewall also offers fine-grained network security controls for interconnected VPCs via the AWS Transit Gateway.

You can also use this to filter your outbound traffic to prevent unwanted data loss, block malware, and satisfy your strict network security compliance requirement. A single AWS Network Firewall can be configured with thousands of rules that can filter out network traffic routed to known bad IP addresses or suspicious domain names. It can also protect the AWS Direct Connect or VPN traffic that originates from client devices and your on-premises environments. The AWS Network Firewall can ensure that only authorized sources of traffic are granted access to your mission-critical VPC resources. It is also capable of performing the same activities as your Intrusion Detection Systems and Intrusion Prevention Systems or IDP/IPS. This is achieved by inspecting all inbound Internet traffic using features such as ACL rules, stateful inspection, protocol detection, intrusion prevention et cetera.

The AWS Network Firewall has 3 basic components, namely the Firewall, the Firewall Policy, and the Rule Group.

A firewall is a resource that you create in AWS Network Firewall. This firewall is connected to the Amazon VPC of your choice, where a network filter will be implemented based on the behavior defined in your firewall policy. Your firewall can have one firewall policy only, which contains rule groups that you define.

You can deploy the firewall in multiple Availability Zones and to one subnet per zone. Each subnet that is associated with your firewall must have at least one available IP address. Afterward, you must update your VPC route tables to send incoming and outgoing traffic through the firewall endpoints.

The second component is the firewall policy. This is a policy that defines the behavior of your firewall using a collection of stateless and stateful rule groups. A firewall policy can be associated with one or more firewalls.



However, a firewall can only have one firewall policy. Changing a firewall policy affects all other firewalls that reference it.

The third component in AWS Network Firewall is called a rule group. This is basically a collection of stateless or stateful rules that define how to inspect and handle the network traffic in your VPC. A rules configuration includes 5-tuple network values and domain name filtering. The 5-tuple format includes the source IP address, source port, destination IP address, destination port, and protocol.

AWS Network Firewall has two types of rules which could be stateless or stateful. The first one is stateless in the sense that it does not have any context of the packet's traffic flow. A stateless rule just checks the packet itself. On the contrary, a stateful rule knows the context or the state of the packet, including its direction flow and other information that is not provided by the packet itself. Each rule in your stateful rule group has an associated order for its evaluation sequence. This concept is similar to network access control lists and security groups. A network ACL is stateless, while a security group is stateful.

You can also choose how you want the AWS Network Firewall to handle the packets that match your rule criteria. You can either pass or drop a packet that was filtered by the network firewall. A packet can also be forwarded to another stateful rule group for re-evaluation. Setting up custom actions is possible as well. A custom action can be configured to publish data to CloudWatch metrics for future network analysis.